



## GDAC-SA PRE-QUALIFICATION RESPONSE

Advanced Analytics Platform  
Saudi Center for Geoscience Data Analytics (GDAC-SA)  
Lithodat Pty Ltd  
ABN: 63 627 008 904  
Submitted: December 2024



هيئة المساحة الجيولوجية السعودية  
Saudi Geological Survey

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# EXECUTIVE SUMMARY

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Lithodat Pty Ltd is honoured to submit our pre-qualification response for the Geological Data Analysis Center (GDAC-SA) AI/Big Data Analytics Platform. We recognize that the Saudi Geological Survey has received approval to establish a transformational platform that will enable Saudi Arabia's Vision 2030 objective of developing the mining sector as the Kingdom's third economic pillar. This platform must integrate diverse geological data types, such as reports, geochemistry, geophysics, remote sensing, geology, and drilling records, from the National Geological Database (NGD), National Information Center (NIC), and new field campaigns, then apply advanced AI analytics to deliver mineral prospectivity predictions, target generation, and exploratory forecasting capabilities. The Saudi Geological Survey correctly identifies that much of this data exists with varying quality, incomplete metadata, inconsistent formats, and aggregations spanning decades of geological work.

We are uniquely positioned to deliver this vision because we have already built exactly what GDAC-SA envisions. Over the past decade, Lithodat together with our government, industry and academic partners, has successfully delivered two national-scale data cleaning and standarding programs of work for federal government organizations across Australia and Canada:

- EarthBank (national geochemistry platform for AuScope Geoscience Australia)
- Isotopes.au (federal stable isotope platform developed with CSIRO and four other Australian government agencies)
- CATCH Database (Canadian Thermochronology Database for Natural Resources Canada).

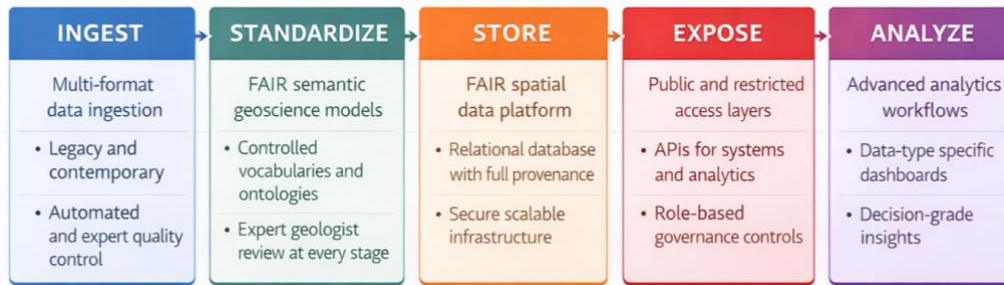
These geological databases integrate the type of multi-agency, multi-decade geological data that GDAC-SA must harmonize, to power the AI-enabled analytics for mineral exploration, environmental assessment, and geoscientific research. Our track record demonstrates the capability, and proven operational delivery of the digital infrastructure GDAC-SA requires.

## The Critical Foundation: Clean Data First for AI Success

The RFQ correctly identifies GDAC-SA's fundamental challenge: creating a unified geospatial analytics platform from data sources with "aggregations of data and gaps, including structured and unstructured data, some of which are without metadata." The Saudi Geological Survey recognizes that Phase 1 must create "the underlying data ecosystem" through data structure design, feature engineering, natural language processing, and new algorithm development. This understanding is precisely correct, and we must emphasize a critical principle that determines whether GDAC-SA succeeds or fails: the only pathway to AI-enabled mineral exploration workflows is through clean, standardized, FAIR (Findable, Accessible, Interoperable and Reusable)-compliant data. Machine learning models cannot overcome poor data quality. No amount of sophisticated AI algorithms can extract reliable mineral prospectivity predictions from inconsistent terminology, missing spatial coordinates, incompatible measurement units, or uncontrolled vocabularies. Attempts to "bolt on" AI to poorly structured data inevitably fail because poor data requires more effort from AI and subsequently produces poor predictions.

Lithodat's "Clean Data First" methodology ensures that every data ingestion, transformation, and storage step creates AI-ready datasets from day one. Our proven five-stage pipeline has successfully powered three national and international data platforms, and this exact architecture will deliver GDAC-SA's requirements:

## End-to-end FAIR data platform enabling AI-ready geoscience



**Figure 1: Lithodat's Clean First Data Pipeline**

**Stage 1 (Data Ingestion)** employs multi-format adapters that handle Excel and CSV datasets, legacy databases, and other unstructured data with automated validation against geoscience vocabularies and quality scoring at point of entry. For the CATCH Database, we digitized 50+ years of legacy data (1970-2021) from 91 disparate sources including analogue paper reports, non-vectorized PDFs, and published figures, georeferencing sample locations from maps without GPS coordinates, exactly the historical data challenge GDAC-SA faces with NGD and NIC archives.

**Stage 2 (Semantic Standardization)** creates SKOS-based controlled vocabularies (ontologies) that map diverse terminology to standardized concepts, enabling AI feature engineering across heterogeneous sources. For Isotopes.au we developed 250+ standardized field mappings validated across six institutional data sources from three federal agencies, directly analogous to GDAC's challenge of harmonizing Saudi Geological Survey data across departments with different conventions. This semantic interoperability is essential for Phase 1's data ecosystem and Phase 2's AI-based mineral system modeling.

**Stage 3 (FAIR-Compliant Database Storage)** implements our comprehensive metadata capture, full audit trail and provenance tracking, and RESTful APIs for programmatic access by AI workflows. Our EarthBank platform achieved CoreTrustSeal certification, the international standard for trustworthy digital repositories, demonstrating our implementation of FAIR (Findable, Accessible, Interoperable, Reusable) data principles. FAIR data is not merely a compliance requirement; it is the technical foundation that enables AI success. Findability through rich metadata allows algorithms to automatically discover relevant features (mineral associations, structural controls, geochemical signatures). Accessibility via RESTful APIs enables external machine learning frameworks (PyTorch, TensorFlow, scikit-learn) to programmatically access training datasets. Interoperability through controlled vocabularies creates the semantic standardization essential for feature engineering. Reusability with comprehensive provenance tracking allows Phase 2 models to be retrained as new data arrives in Phases 3-4 without quality degradation.

**Stage 4 (SGS Live Portal Development)** creates dual-access systems: public-facing data portals for open geoscience access by researchers and industry, plus private secure areas for SGS internal workflows with role-based access control and user management. EarthBank and Isotopes.au are fully operational public platforms serving geoscientists globally while maintaining secure administrative features for government analytical requirements.

**Stage 5 (AI/ML Application Layer)** delivers advanced analytics through feature engineering from clean structured data, machine learning model training for classification /regression/clustering, mineral prospectivity mapping and target generation, and time series analysis for trend detection.

# OVERVIEW

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Table 1: Company Overview

|                       |                                       |
|-----------------------|---------------------------------------|
| RFQ Reference Number: | 251140007625                          |
| Government Entity:    | Saudi Geological Survey (SGS)         |
| Applicant:            | Lithodat Pty Ltd                      |
| ABN:                  | 63 627 008 904                        |
| Submission Date:      | December 2025                         |
| Submission Platform:  | Etimad Portal                         |
| Prepared By:          | Keith Dimech, Chief Operating Officer |
| Contact Email:        | keith.dimech@lithodat.com             |
| Contact Phone:        | +61 428 971 396                       |

**Currency Note:** All financial values shown in Australian Dollars (AUD) and Saudi Riyals (SAR) at exchange rate: AUD 1.00 = SAR 2.45 (Monday 1st of December).

# APPLICANT INFORMATION

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## Application Information Form

Table 2: Advanced

|                                       |                                                                    |
|---------------------------------------|--------------------------------------------------------------------|
| <b>Name of Applicant</b>              | Lithodat Pty Ltd                                                   |
| <b>Commercial Registration Number</b> | ABN 63 627 008 904 (Australian Business Number)                    |
| <b>Establishment Date</b>             | 25 June 2018                                                       |
| <b>Years in Operation</b>             | 7 years (2018-2025)                                                |
| <b>Company Type</b>                   | Proprietary Limited Company (Pty Ltd)                              |
| <b>Function</b>                       | Geological Artificial Intelligence, Database and Software Services |
| <b>Person Responsible</b>             | Fabian Kohlmann                                                    |
| <b>Capital Value<sup>1</sup></b>      | AUD \$296,608 / SAR 726,690                                        |

<sup>1</sup> Capital Value = Net Assets as at 30 June 2025 per FY2025 Annual Report (Total Assets \$430,819 - Total Liabilities \$134,211 = \$296,608)

## Contact Information

Table 3: Address

|                       |                              |
|-----------------------|------------------------------|
| <b>Street Address</b> | 94 Stephensons Road          |
| <b>City</b>           | Mount Waverley, Victoria     |
| <b>Zip Code</b>       | 3149                         |
| <b>Country</b>        | Australia                    |
| <b>P.O. Box</b>       | N/A                          |
| <b>Telephone</b>      | +61 481 977 499              |
| <b>Email</b>          | fabian.kohlmann@lithodat.com |
| <b>Website</b>        | www.lithodat.com             |

|                       |                                                                       |
|-----------------------|-----------------------------------------------------------------------|
| <b>Site (Offices)</b> | Melbourne, Australia<br><br>Dublin, Ireland<br><br>Hermosillo, Mexico |
|-----------------------|-----------------------------------------------------------------------|

Company Ownership Structure

Source: Appendix A - ASIC Company Extract (02-Dec-2025)  
Total Class A Shares: 1,000

Table 4: Company Ownership

| # | Owner/Partner Name       | Percentage         | Nationality         |
|---|--------------------------|--------------------|---------------------|
| 1 | Dr. Fabian Kohlmann      | 50.1% (501 shares) | German / Australian |
| 2 | Gerd Moritz Theile       | 24.0% (240 shares) | German              |
| 3 | Dr. Wayne Peter Noble    | 14.9% (149 shares) | Australian          |
| 4 | Nilesh Ambadas Vyavahare | 8.0% (80 shares)   | British / Indian    |
| 5 | Romain Beucher           | 3.0% (30 shares)   | French / Australian |

Company Representative Information

Table 5:Company Representative

|                            |                               |
|----------------------------|-------------------------------|
| Name of Responsible Person | Fabian Kohlmann               |
| Job Title                  | Chief Executive Officer (CEO) |
| Telephone                  | +61 481 977 499               |
| Mobile                     | +61 481 977 499               |
| Email                      | fabian.kohlmann@lithodat.com  |

# TECHNICAL AND ADMINISTRATIVE CAPABILITIES

## Previous Work Experience

**Table 6: Previous Work Experience**

|                                                             |                                                                                                                                                   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of years of experience in the field of project       | Lithodat - 7 years (since 2018)                                                                                                                   |
| Number of similar projects executed in the last three years | 3 major projects<br>Appendix G -Earthbank – Project Summary<br>Appendix H - Isotopes.au – Project Summary<br>Appendix I - NRCan – Project Summary |

## Key Performance Outcomes

**Table 7: Performance Metrics**

| Performance Metric  | EarthBank | Isotopes-AU | NRCan     |
|---------------------|-----------|-------------|-----------|
| On-Time Delivery    | Excellent | Excellent   | Excellent |
| Technical Quality   | Excellent | Excellent   | Excellent |
| Client Satisfaction | Excellent | Excellent   | Excellent |
| Budget Compliance   | Excellent | Excellent   | Excellent |

## Reference Contacts

**Table 8: Reference Contacts**

| Project     | Reference Contact   | Title                                     | Email                           |
|-------------|---------------------|-------------------------------------------|---------------------------------|
| EarthBank   | Prof. Brent McInnes | AuScope Earthbank Director                | b.mcinnnes@curtin.edu.au        |
| Isotopes-AU | Dr. Nina Welti      | CSIRO – Environmental Isotopes Laboratory | nina.welti@csiro.au             |
| NRCan       | Dr. Jeremy Powell   | Geological Survey of Canada (NRCan)       | jeremy.powell@nrcan-rncan.gc.ca |

## Quality

**Table 9: Quality Assurance**

|                                    |                                                        |
|------------------------------------|--------------------------------------------------------|
| <b>Quality Assurance Standards</b> | Quality Management System for Geological Data Services |
| <b>Evidence Provided</b>           | Quality Policy (CEO-signed)                            |

## Health Safety and Environment

**Table 10: HSE Assurance**

|                          |                                  |
|--------------------------|----------------------------------|
| <b>HSE Standards</b>     | Lithodat HSE Policy              |
| <b>Evidence Provided</b> | HSE Policy Document (CEO-signed) |

## Existing Contractual Obligations

**Table 11: Existing Contractual Obligations**

|                                    |                                                      |
|------------------------------------|------------------------------------------------------|
| <b>Number of existing projects</b> | 2 active projects, 1 in Negotiation (beginning 2026) |
| <b>Value of existing projects</b>  | AUD \$3,800,000 combined<br>SAR \$9,310,000 combined |
| <b>Status</b>                      | Both projects ongoing and in good standing           |

## Human Resources

**Table 12: Human Resources**

|                                      |                                                  |
|--------------------------------------|--------------------------------------------------|
| <b>Total Number of Employees</b>     | 19 (3 Management + 10 Full-time + 6 Contractors) |
| <b>Number of Saudi Employees</b>     | 1 (Dr. Mahdi AbuAli - GDAC Saudi Lead)           |
| <b>Percentage of Saudi Employees</b> | 5.3%                                             |

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## ADMINISTRATIVE STAFF EXPERIENCE

Table 13: Project Administration and Leadership

| # | Name                | Function                  | Specialization / Area of Expertise                                                                                                                                           | Years     |
|---|---------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1 | Dr. Fabian Kohlmann | Managing Director and CEO | Geoscience Leadership<br>Strategic Planning, Business Development<br>Former Halliburton Thermochemistry Product Lead<br>Geoscience Data Management<br>FAIR Data Architecture | 15+ years |
| 2 | Dr. Mahdi AbuAli    | GDAC Saudi Lead           | Global Geology Leader<br>Saudi Government Relations,<br>Saudi Aramco (35+ years),<br>Executive Leadership,<br>Petroleum Systems                                              | 35+ years |
| 3 | Keith Dimech        | Chief Operating Officer   | Project Management,<br>Financial Operations,<br>Human Relations<br>Legal                                                                                                     | 15+ years |
| 4 | Juan Baca           | Quality Manager           | Data Quality Management<br>Geology Process Leadership<br>Database Operations<br>Training                                                                                     | 10+ years |
| 5 | Vinko Novak         | Head of Data Security     | Data Security<br>Information Protection<br>Compliance                                                                                                                        | 20+ years |

| # | Name            | Function               | Specialization / Area of Expertise                             | Years     |
|---|-----------------|------------------------|----------------------------------------------------------------|-----------|
|   |                 |                        | Security Architecture                                          |           |
| 6 | Annemarie Grass | Head of Cyber Security | Cyber Security<br>ISO 27001 Accreditation<br>Incident Response | 15+ Years |

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# PROFESSIONAL STAFF EXPERIENCE

Table 14: Senior Technical Team

| # | Name                  | Function                           | Specialization / Area of Expertise                                                                                                                                        | Years     |
|---|-----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1 | Dr. Wayne Peter Noble | Chief Information Officer          | PhD – Thermochronology<br>Software development:<br>Enterprise Architecture,<br>Cloud Systems,<br>Agile Leadership,<br>Software Development,<br>Continuous Delivery Expert | 25+ years |
| 2 | Dr. Qusay Abeed       | GDAC Geological Technical Director | Geology & Geochemistry,<br>Basin Modelling,<br>CO2 Storage,<br>Native Arabic Speaker,<br>C++ Development,<br>Quality Assurance Expert,<br>Delivery Lead                   | 18+ years |
| 3 | Gerd Moritz Theile    | Chief Technology Officer           | PostgreSQL Architecture,<br>Database Design,<br>LithoSurfer Platform,<br>Java Development                                                                                 | 25+ years |
| 4 | Dr. Behnam Sadeghi    | ML Technical Advisor               | ML/AI, Geospatial AI,<br>Python Package Developer<br>2023 IAMG Vistelius Award<br>CSIRO Research Fellow<br>58+ Publications                                               | 15+ years |

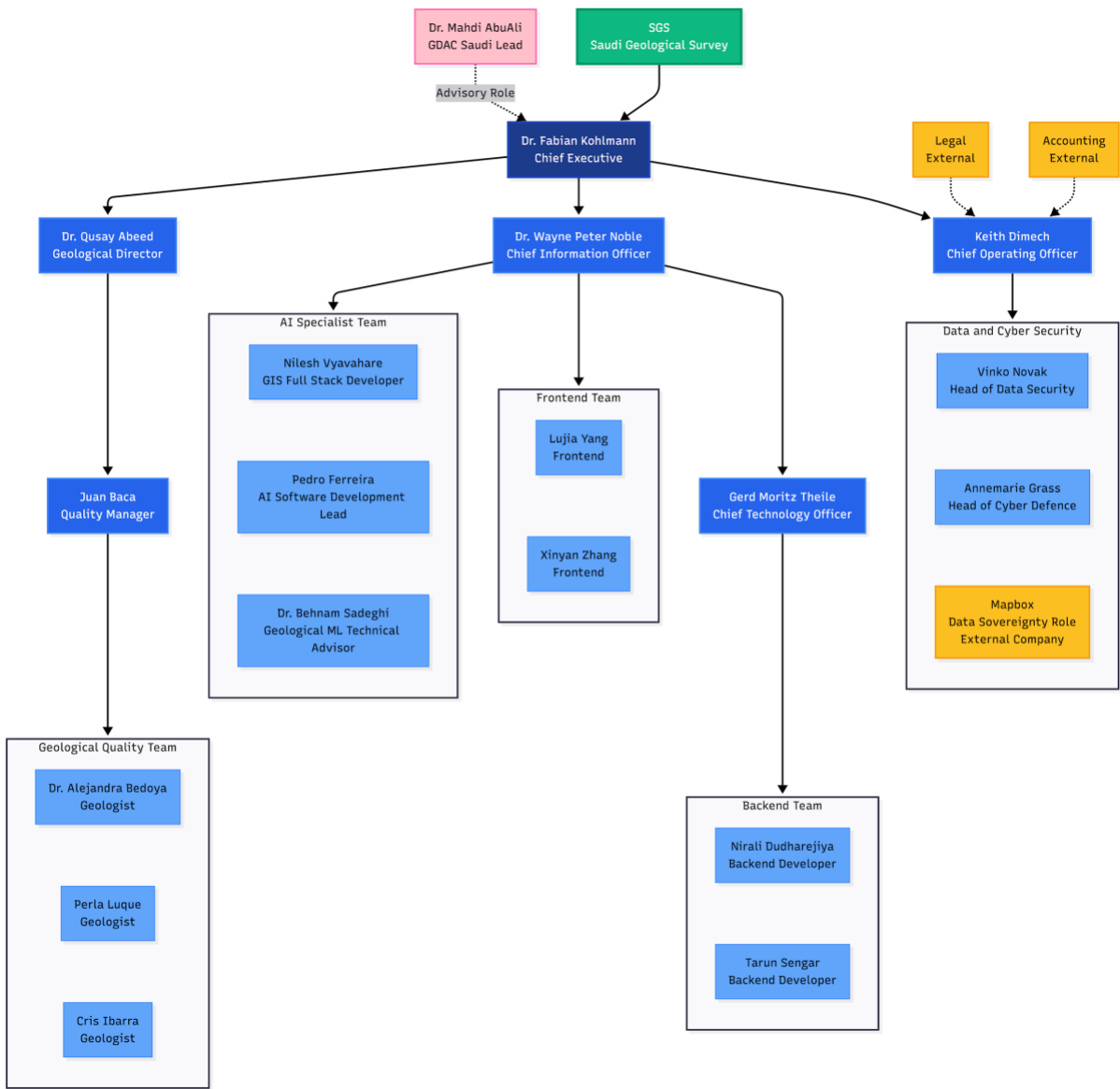
| # | Name             | Function                     | Specialization / Area of Expertise                                                                                                                                                                                                                                    | Years       |
|---|------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|   |                  |                              | Author on Fractals/Multifractals<br>Fulbright Scholar                                                                                                                                                                                                                 |             |
| 5 | Pedro Ferreira   | AI Software Development Lead | Technology and AI/ML Strategy, Enterprise AI Implementation, Power Apps AI Builder                                                                                                                                                                                    | 10-15 years |
| 6 | Nilesh Vyavahare | GIS Full Stack Developer     | Senior Full Stack Engineer (GIS) at<br><br>Principal Software Developer at Neftex - 5+ years (geological applications)<br><br>Earth Observation & satellite data processing<br><br>Geospatial ETL workflows (FME, Apache Airflow)<br><br>ArcGIS development & plugins | 18 + years  |

**Table 15: Junior Technical Team**

| # | Name               | Function                        | Specialization / Area of Expertise                                | Years    |
|---|--------------------|---------------------------------|-------------------------------------------------------------------|----------|
| 1 | Xinyan Zhang       | Frontend Developer              | React, TypeScript, UI/UX Implementation, Responsive Design        | 3+ years |
| 2 | Lujia Yang         | Frontend Developer              | React, TypeScript, UI/UX Implementation, Data Science (MSc)       | 1+ years |
| 3 | Tarun Sengar       | Backend Developer               | Java, AWS, Backend Systems, API Development                       | 5+ years |
| 4 | Nirali Dudharejiya | Backend Developer               | Java, Liquibase, Backend Systems, Database Management             | 5+ years |
| 5 | Cris Ibarra        | Geology Data Quality Specialist | Data Entry, Data Validation, Quality Control, Database Operations | 3+ years |

| # | Name                 | Function                   | Specialization / Area of Expertise                                | Years     |
|---|----------------------|----------------------------|-------------------------------------------------------------------|-----------|
| 6 | Perla Luque          | Geology Quality Specialist | Data Entry, Data Validation, Quality Control, Database Operations | 3+ years  |
| 7 | Dr. Alejandra Bedoya | Geology Quality Specialist | Geochronology, Thermochronology, Tectonics, Data Validation       | 10+ years |

# PROJECT ORGANISATION CHART



## EXPERIENCE IN SIMILAR PROJECTS

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### Project 1: EarthBank - National Geochronology Archive

Table 16: Earthbank

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b>               | EarthBank – Global Geochemistry Data Platform                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Full Details</b>               | Appendix G Earthbank – Project Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Project Link</b>               | <a href="https://app.ausgeochem.org/">https://app.ausgeochem.org/</a><br><a href="https://www.auscope.org.au">https://www.auscope.org.au</a>                                                                                                                                                                                                                                                                                                                                               |
| <b>Project Location</b>           | Australia and Global                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Project Content</b>            | <p>Geological data platform integrating geochemistry and geochronology data from multiple universities, museums and research institutions across Australia. Advanced analytics for mineral exploration and tectonic history reconstruction. The platform manages over 350,000 academically verified and quality checked samples .</p> <p>The database serves national research infrastructure and mineral exploration initiatives internationally and across the Australian continent.</p> |
| <b>Project Owner</b>              | AuScope / NCRIS (National Collaborative Research Infrastructure Strategy)                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Contract Value</b>             | AUD \$2,900,000 / SAR 7,105,000                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Contract Duration</b>          | 5+ years (2020 - Present, ongoing)                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Start Date</b>                 | 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Delivery Date</b>              | Phase 1 Delivered 2023; Ongoing enhancements                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Name of Person Responsible</b> | <a href="#">Dr. Fabian Kohlmann</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Contact Information</b>        | Dr Brett McInnes                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Email</b>                      | b.mcinnes@curtin.edu.au                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Project 2: Isotopes.au - National Isotopic Data Platform

Table 17: Isotopes.au

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b>               | Isotopes.au - National Isotopic Data Platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Full Details</b>               | Appendix H - Isotopes.au – Project Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Project Location</b>           | Australia (National)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Project Link</b>               | <a href="https://isotopes.au/">https://isotopes.au/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Project Content</b>            | <p>Platform for managing and analysing federal stable isotope data across geoscience disciplines. Integration of datasets from CSIRO, ANSTO, Geoscience Australia, and National Measurement Institute. The platform includes advanced data standardization, quality control workflows, and isotopic ratio visualization tools.</p> <p>Registered with AusIndustry R&amp;D program (IISA0053835) demonstrating innovation excellence. Supporting critical minerals exploration and environmental geoscience research across Australia.</p> |
| <b>Project Owner</b>              | CSIRO / ANSTO / Geoscience Australia / NMI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Contract Value</b>             | AUD \$400,000 / SAR 980,000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Contract Duration</b>          | 4.5 years (July 2024 - December 2028)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Start Date</b>                 | July 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Delivery Date</b>              | December 2028 (in progress, on schedule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Name of Person Responsible</b> | Dr. Wayne Noble                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Client Contact Information</b> | Dr Nina Welti - CSIRO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Email</b>                      | nina.welti@csiro.au                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## Project 3: Canadian ThermoCHronology (CATCH) Database - Northern Canada

Table 18: NRCan CATCH Database

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b>               | Natural Resources Canada Geological Data Project                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Project Location</b>           | Canada (National)                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Full Details</b>               | Appendix I - NRCan – Project Summary                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Project Site</b>               | <a href="https://app.lithodat.com/">https://app.lithodat.com/</a>                                                                                                                                                                                                                                                                                                                                                               |
| <b>Project Content</b>            | <p>Development and delivery of Canada's national low-temperature thermochronology database for Natural Resources Canada's Geological Survey of Canada (GSC). The CATCH (Canadian ThermoCHronology) database is a comprehensive, FAIR-compliant (Findable, Accessible, Interoperable, Reuseable) platform for accessing Canadian low-temperature thermochronology (LTT) data.</p> <p>All Data available through Lithosurfer.</p> |
| <b>Project Owner</b>              | Natural Resources Canada (NRCan) - Canadian Federal Government                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Contract Value</b>             | AUD \$400,000 / SAR 980,000                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Contract Duration</b>          | 1.5 years (October 2021 - March 2023)                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Start Date</b>                 | October 2021                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Delivery Date</b>              | March 2023 (DELIVERED)                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Name of Person Responsible</b> | <a href="#">Dr. Fabian Kohlmann</a>                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Client Contact Information</b> | <p>Geological Survey of Canada (NRCan)</p> <p>Dr. Jeremy Powell</p>                                                                                                                                                                                                                                                                                                                                                             |
| <b>Email</b>                      | <a href="mailto:jeremy.powell@nrcan-rncan.gc.ca">jeremy.powell@nrcan-rncan.gc.ca</a>                                                                                                                                                                                                                                                                                                                                            |

# FINANCIAL CAPACITY CRITERIA

## 6.1 Balance Sheet Statement (Last 3 Financial Years)

Table 19: Balance Sheet Statement

| Balance Sheet Line Item   | FY2025                         | FY2024                       | FY2023                       |
|---------------------------|--------------------------------|------------------------------|------------------------------|
| Total Assets              | AUD \$430,819<br>SAR 1,055,506 | AUD \$216,906<br>SAR 531,420 | AUD \$130,477<br>SAR 319,669 |
| Current Assets            | AUD \$364,840<br>SAR 893,858   | AUD \$176,800<br>SAR 433,160 | AUD \$93,690<br>SAR 229,541  |
| Cash and Cash Equivalents | AUD \$274,855<br>SAR 673,395   | AUD \$100,406<br>SAR 245,995 | AUD \$40,605<br>SAR 99,482   |
| Accounts Receivable       | AUD \$17,487<br>SAR 42,843     | AUD \$76,394<br>SAR 187,165  | AUD \$53,001<br>SAR 129,852  |
| Total Liabilities         | AUD \$134,211<br>SAR 328,817   | AUD \$80,012<br>SAR 196,029  | AUD \$73,574<br>SAR 180,256  |
| Current Liabilities       | AUD \$60,404<br>SAR 147,990    | AUD \$40,106<br>SAR 98,260   | AUD \$36,787<br>SAR 90,128   |
| Net Assets (Total Equity) | AUD \$296,608<br>SAR 726,690   | AUD \$136,894<br>SAR 335,390 | AUD \$56,903<br>SAR 139,412  |

## 6.2 Financial Performance Indicators

Table 20: Financial Performance Indicators

| Financial Ratio       | Formula                                    | FY2025 | FY2024 | FY2023 |
|-----------------------|--------------------------------------------|--------|--------|--------|
| Cash Ratio            | Cash / Current Liabilities                 | 4.55   | 2.50   | 1.10   |
| Current Ratio         | Current Assets / Current Liabilities       | 6.04   | 4.41   | 2.55   |
| Quick Liquidity Ratio | (Cash + Receivables) / Current Liabilities | 4.84   | 4.41   | 2.54   |

# DECLARATION

I, the undersigned, declare that:

1. All information provided in this qualification application is true, accurate, and complete to the best of my knowledge.
2. All attached documents and certificates are valid and authentic.
3. Lithodat Pty Ltd agrees to comply with all applicable laws, regulations, and requirements of the Kingdom of Saudi Arabia.
4. We understand that any false or misleading information may result in disqualification from this tender process.
5. We agree to maintain confidentiality of any information obtained during the qualification process.
6. We acknowledge that the Saudi Geological Survey reserves the right to modify dates, requirements, or cancel the RFQ process at any time.
7. If selected, we commit to registering in the Kingdom of Saudi Arabia and complying with all in-Kingdom requirements including Saudization targets.

**Authorized Signatory:** Dr. Fabian Kohlmann

**Position:** Managing Director and CEO

**Company:** Lithodat Pty Ltd

**Date:**

**Signature:**

## LITHODAT PTY LTD

ABN 63 627 008 904

94 Stephensons Road, Mount Waverley, Victoria 3149, Australia

Email: [fabian.kohlmann@lithodat.com](mailto:fabian.kohlmann@lithodat.com) | Website: [www.lithodat.com](http://www.lithodat.com)

Phone: +61 481 977 499

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## **Appendix A**

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**Australian Security and Investment Commission – Lithodat Pty Ltd**

## Appendix B

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Financial Statement FY2024-2025

# Annual Report

Lithodat Pty Ltd  
ABN 63 627 008 904  
For the year ended 30 June 2025

Prepared by Scott Partners Chartered Accountants

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| 4 | Directors Declaration             |
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| 7 | Balance Sheet                     |
| 9 | Notes to the Financial Statements |

# Compilation Report

## Lithodat Pty Ltd For the year ended 30 June 2025

Compilation report to Lithodat Pty Ltd

We have compiled the accompanying special purpose financial statements of Lithodat Pty Ltd, which comprise the balance sheet as at 30 June 2025, the income statement, a summary of significant accounting policies and other explanatory notes. The specific purpose for which the special purpose financial statements have been prepared is set out in Note 1.

### The Responsibility of the Directors

The directors of Lithodat Pty Ltd are solely responsible for the information contained in the special purpose financial statements, the reliability, accuracy and completeness of the information and for the determination that the basis of accounting used is appropriate to meet their needs and for the purpose that financial statements were prepared.

### Our Responsibility

On the basis of information provided by the directors we have compiled the accompanying special purpose financial statements in accordance with the basis of accounting as described in Note 1 to the financial statements and APES 315 *Compilation of Financial Information*.

We have applied our expertise in accounting and financial reporting to compile these financial statements in accordance with the basis of accounting described in Note 1 to the financial statements. We have complied with the relevant ethical requirements of APES 110 *Code of Ethics for Professional Accountants*.

### Assurance Disclaimer

Since a compilation engagement is not an assurance engagement, we are not required to verify the reliability, accuracy or completeness of the information provided to us by management to compile these financial statements. Accordingly, we do not express an audit opinion or a review conclusion on these financial statements.

The special purpose financial statements were compiled exclusively for the benefit of the directors who are responsible for the reliability, accuracy and completeness of the information used to compile them. We do not accept responsibility for the contents of the special purpose financial statements.



Donald Shaw  
**Scott Partners Pty Ltd**  
Level 2, Suite 1, 2-6 Glenferrie Rd  
Malvern VIC 3144

15 October 2025 | 10:55 AEDT

Dated this ..... day of ..... 2025

# Directors Declaration

## Lithodat Pty Ltd For the year ended 30 June 2025

The directors have determined that the company is not a reporting entity and that this special purpose financial report should be prepared in accordance with the accounting policies outlined in Note 1 to the financial statements.

The directors of the company declare that:

- 1. The financial statements and notes, present fairly the company’s financial position as at 30 June 2025 and its performance for the year ended on that date in accordance with the accounting policies described in Note 1 to the financial statements; and
- 2. In the directors’ opinion there are reasonable grounds to believe that the company will be able to pay its debts as and when they become due and payable.

This declaration is made in accordance with a resolution of the Board of Directors.

Director: Fabian Kohlmann Fabian Kohlmann

Director: Gerd Moritz Walter Theile M. Theile

Director: Wayne Noble Wayne Noble

17 October 2025 | 00:48 EDT  
Dated this ..... day of ..... 2025

# Income Statement

## Lithodat Pty Ltd

For the year ended 30 June 2025

|                                              | NOTES | 2025             | 2024           |
|----------------------------------------------|-------|------------------|----------------|
| <b>Income</b>                                |       |                  |                |
| Sales                                        |       | 1,192,265        | 676,285        |
| <b>Total Income</b>                          |       | <b>1,192,265</b> | <b>676,285</b> |
| <b>Other Income</b>                          |       |                  |                |
| Foreign Currency Gains and Losses            |       | 29,637           | (2,240)        |
| Interest Income                              |       | 2,112            | 296            |
| Government Grant                             |       | 20,000           | 56,000         |
| Other Revenue                                |       | 847              | 2,159          |
| Research & Development Offset Income         |       | 212,233          | 84,120         |
| Unrealised Foreign Currency Gains and Losses |       | 41               | -              |
| <b>Total Other Income</b>                    |       | <b>264,870</b>   | <b>140,335</b> |
| <b>Total Income</b>                          |       | <b>1,457,135</b> | <b>816,620</b> |
| <b>Expenses</b>                              |       |                  |                |
| Accounting                                   |       | 5,139            | 4,503          |
| Advertising                                  |       | 5,923            | 6,827          |
| Bank Fees                                    |       | 5,652            | 1,207          |
| Borrowing Expense                            |       | 595              | -              |
| Conferences & Trade Fairs                    |       | 57,210           | 44,026         |
| Consulting                                   |       | 384,543          | 166,009        |
| Depreciation                                 |       | 3,127            | 10,169         |
| Donations                                    |       | 100              | -              |
| Entertainment                                |       | 2,890            | 2,018          |
| Filing Fees                                  |       | 321              | 310            |
| Freight & Courier                            |       | 5,718            | 260            |
| General Expenses                             |       | 2,181            | 3,458          |
| Gift to Clients                              |       | 678              | 630            |
| Insurance                                    |       | 20,383           | 22,811         |
| Interest Expense                             |       | 6,406            | 5,351          |
| Legal expenses                               |       | 4,100            | 5,075          |
| Memberships                                  |       | 8,870            | 8,166          |
| Motor Vehicle                                |       | 3,208            | 123            |
| Non-deductible expense                       |       | -                | 1,139          |
| Office Expenses                              |       | 5,487            | 1,957          |
| Printing & Stationery                        |       | 295              | 75             |
| Subscriptions                                |       | 59,100           | 27,906         |
| Superannuation                               |       | 47,530           | 27,793         |
| Telephone & Internet                         |       | 3,507            | 2,906          |
| Travel and Accommodation                     |       | 101,032          | 68,153         |
| Wages and Salaries                           |       | 413,304          | 252,664        |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

|                                 | NOTES | 2025      | 2024    |
|---------------------------------|-------|-----------|---------|
| Website expenses                |       | 1,850     | 7,061   |
| Total Expenses                  |       | 1,149,152 | 670,595 |
| Profit/(Loss) before Taxation   |       | 307,983   | 146,024 |
| Income Tax Expense              |       |           |         |
| Income Tax Expense              |       | 148,069   | 66,233  |
| Total Income Tax Expense        |       | 148,069   | 66,233  |
| Net Profit After Tax            |       | 159,914   | 79,791  |
| Net Profit After Dividends Paid |       | 159,914   | 79,791  |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

# Balance Sheet

## Lithodat Pty Ltd As at 30 June 2025

|                                                          | NOTES | 30 JUNE 2025   | 30 JUNE 2024   |
|----------------------------------------------------------|-------|----------------|----------------|
| <b>Assets</b>                                            |       |                |                |
| <b>Current Assets</b>                                    |       |                |                |
| Cash on Hand                                             |       | 1,000          | 1,000          |
| Accounts Receivable                                      |       | 17,487         | 125,400        |
| <b>Bank Accounts</b>                                     |       |                |                |
| American Express Platinum Busi                           |       | (66,956)       | (46,938)       |
| CBA Business Account                                     |       | 96,781         | 65,733         |
| CBA Business Online Saver                                |       | 10             | 10             |
| CBA Capital Growth Account                               |       | 252,348        | 20,236         |
| Wise Euro                                                |       | 7              | 38             |
| <b>Total Bank Accounts</b>                               |       | <b>282,189</b> | <b>39,078</b>  |
| Income Tax Refund                                        |       | 64,164         | 17,887         |
| <b>Total Current Assets</b>                              |       | <b>364,840</b> | <b>183,365</b> |
| <b>Non-Current Assets</b>                                |       |                |                |
| Trademarks                                               |       | 11,323         | 11,323         |
| Computer Equipment                                       |       | 15,805         | 15,805         |
| Less Accumulated Depreciation on Computer Equipment      |       | (15,805)       | (15,805)       |
| Motor Vehicle                                            |       | 57,784         | -              |
| Less Accumulated Depreciation on Motor Vehicle           |       | (3,127)        | -              |
| <b>Total Non-Current Assets</b>                          |       | <b>65,980</b>  | <b>11,323</b>  |
| <b>Total Assets</b>                                      |       | <b>430,819</b> | <b>194,687</b> |
| <b>Liabilities</b>                                       |       |                |                |
| <b>Current Liabilities</b>                               |       |                |                |
| Child Support Payable                                    |       | 517            | 1,099          |
| GST                                                      |       | 16,437         | 19,159         |
| PAYG Withholdings Payable                                |       | 28,854         | 18,924         |
| Superannuation Payable                                   |       | 14,595         | 8,812          |
| <b>Total Current Liabilities</b>                         |       | <b>60,404</b>  | <b>47,993</b>  |
| <b>Non-Current Liabilities</b>                           |       |                |                |
| <b>Hire Purchase Agreements</b>                          |       |                |                |
| CBA Asset Finance (Current) - Tesla Model Y              |       | 15,255         | -              |
| Unexpired Interest Charges (Current) - Tesla Model Y     |       | (3,935)        | -              |
| CBA Asset Finance (Non-Current) - Tesla Model Y          |       | 54,662         | -              |
| Unexpired Interest Charges (Non-Current) - Tesla Model Y |       | (6,628)        | -              |
| <b>Total Hire Purchase Agreements</b>                    |       | <b>59,353</b>  | <b>-</b>       |
| Loan - Kristy Kohlmann                                   |       | 14,454         | 10,000         |
| <b>Total Non-Current Liabilities</b>                     |       | <b>73,807</b>  | <b>10,000</b>  |
| <b>Total Liabilities</b>                                 |       | <b>134,211</b> | <b>57,993</b>  |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

|                   | NOTES | 30 JUNE 2025 | 30 JUNE 2024 |
|-------------------|-------|--------------|--------------|
| Net Assets        |       | 296,608      | 136,694      |
| Equity            |       |              |              |
| Retained Earnings |       | 295,608      | 135,694      |
| Share Capital     |       | 1,000        | 1,000        |
| Total Equity      |       | 296,608      | 136,694      |

# Notes to the Financial Statements

## Lithodat Pty Ltd

### For the year ended 30 June 2025

#### 1. Statement of Significant Accounting Policies

The directors have determined that the company is not a reporting entity and accordingly, this financial report is a special purpose report prepared for the sole purpose of distributing a financial report to members and must not be used for any other purpose. The directors have determined that the accounting policies adopted are appropriate to meet the needs of the members.

The financial report has been prepared on an accrual basis and under the historical cost convention, except for certain assets, which, as noted, have been written down to fair value as a result of impairment. Unless otherwise stated, the accounting policies adopted are consistent with those of the prior year.

The accounting policies that have been adopted in the preparation of the statements are as follows:

##### Income Tax

The income tax expense for the year comprises current income tax expense. The company does not apply deferred tax. Current income tax expense charged to the profit or loss is the tax payable on taxable income calculated using applicable income tax rates enacted, or substantially enacted, as at 30 June 2025. Current tax liabilities are therefore measured at the amounts expected to be paid to the relevant taxation authority.

##### Property, Plant and Equipment

Property, plant and equipment is initially recorded at the cost of acquisition or fair value less, if applicable, any accumulated depreciation and impairment losses. Plant and equipment that has been contributed at no cost, or for nominal cost, is valued and recognised at the fair value of the asset at the date it is acquired. The plant and equipment is reviewed annually by directors to ensure that the carrying amount is not in excess of the recoverable amount from these assets. The recoverable amount is assessed on the basis of the expected net cash flows that will be received from the utilisation of the assets and the subsequent disposal. The expected net cash flows have been discounted to their present values in estimating recoverable amounts.

Freehold land and buildings are measured at their fair value, based on periodic, but at least triennial, valuations by independent external valuers, less subsequent depreciation for buildings.

Increases in the carrying amount of land and buildings arising on revaluation are credited in equity to a revaluation surplus. Decreases against previous increases of the same asset are charged against fair value reserves in equity. All other decreases are charged to profit or loss.

Any accumulated depreciation at the date of revaluation is offset against the gross carrying amount of the asset and the net amount is restated to the revalued amount of the asset.

##### Intangibles

Goodwill is recognised as the excess of the purchase price for a business acquired over the fair value of the net assets at the date of acquisition. Goodwill is assessed for impairment annually and is carried at cost less accumulated impairment losses.

##### Cash and Cash Equivalents

Cash and cash equivalents include cash on hand, deposits held on call with banks, other short-term highly liquid investments with original maturities of three months or less, and bank overdrafts.

##### Revenue Recognition

Revenue from the sale of goods is recognised upon the delivery of goods to customers.

Revenue from the rendering of services is recognised upon the delivery of the services to customers.

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These notes should be read in conjunction with the attached compilation report.

Revenue from commissions is recognised upon delivery of services to customers.  
Revenue from interest is recognised using the effective interest rate method.  
Revenue from dividends is recognised when the entity has a right to receive the dividend.

All revenue is stated net of the amount of goods and services tax (GST).

### **Goods and Services Tax**

Transactions are recognised net of GST, except where the amount of GST incurred is not recoverable from the Australian Taxation Office (ATO).

Receivables and payables are stated inclusive of the amount of GST receivable or payable. The net amount of GST recoverable from, or payable to, the ATO is included with other receivables or payables in the balance sheet.

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These notes should be read in conjunction with the attached compilation report.

## Appendix C

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Financial Statement FY2023-2024

# Annual Report

Lithodat Pty Ltd  
ABN 63 627 008 904  
For the year ended 30 June 2024

Prepared by Scott Partners Chartered Accountants

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# Compilation Report

## Lithodat Pty Ltd

## For the year ended 30 June 2024

Compilation report to Lithodat Pty Ltd

We have compiled the accompanying special purpose financial statements of Lithodat Pty Ltd, which comprise the balance sheet as at 30 June 2024, the income statement, a summary of significant accounting policies and other explanatory notes. The specific purpose for which the special purpose financial statements have been prepared is set out in Note 1.

### The Responsibility of the Directors

The directors of Lithodat Pty Ltd are solely responsible for the information contained in the special purpose financial statements, the reliability, accuracy and completeness of the information and for the determination that the basis of accounting used is appropriate to meet their needs and for the purpose that financial statements were prepared.

### Our Responsibility

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We have applied our expertise in accounting and financial reporting to compile these financial statements in accordance with the basis of accounting described in Note 1 to the financial statements. We have complied with the relevant ethical requirements of APES 110 *Code of Ethics for Professional Accountants*.

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The special purpose financial statements were compiled exclusively for the benefit of the directors who are responsible for the reliability, accuracy and completeness of the information used to compile them. We do not accept responsibility for the contents of the special purpose financial statements.

*Donald Shaw*

Donald Shaw  
**Scott Partners Pty Ltd**  
Level 2, Suite 1, 2-6 Glenferrie Rd  
Malvern VIC 3144

19 August 2024 | 12:57 AEST

Dated this ..... day of ..... 2024

# Directors Declaration

## Lithodat Pty Ltd For the year ended 30 June 2024

The directors have determined that the company is not a reporting entity and that this special purpose financial report should be prepared in accordance with the accounting policies outlined in Note 1 to the financial statements.

The directors of the company declare that:

- 1. The financial statements and notes, present fairly the company’s financial position as at 30 June 2024 and its performance for the year ended on that date in accordance with the accounting policies described in Note 1 to the financial statements; and
- 2. In the directors’ opinion there are reasonable grounds to believe that the company will be able to pay its debts as and when they become due and payable.

This declaration is made in accordance with a resolution of the Board of Directors.

Director: Fabian Kohlmann Fabian Kohlmann

Director: Gerd Moritz Walter Theile M. Theile

Director: Wayne Noble Wayne Noble

19 August 2024 | 18:06 AEST

Dated this ..... day of ..... 2024

# Income Statement

## Lithodat Pty Ltd

For the year ended 30 June 2024

|                                      | NOTES | 2024           | 2023           |
|--------------------------------------|-------|----------------|----------------|
| <b>Income</b>                        |       |                |                |
| Sales                                |       | 676,285        | 450,125        |
| <b>Total Income</b>                  |       | <b>676,285</b> | <b>450,125</b> |
| <b>Other Income</b>                  |       |                |                |
| Interest Income                      |       | 296            | 84             |
| Government Grant                     |       | 56,000         | -              |
| Other Revenue                        |       | 2,159          | -              |
| Research & Development Offset Income |       | 84,120         | 78,278         |
| <b>Total Other Income</b>            |       | <b>142,575</b> | <b>78,362</b>  |
| <b>Total Income</b>                  |       | <b>818,860</b> | <b>528,487</b> |
| <b>Expenses</b>                      |       |                |                |
| Accounting                           |       | 4,503          | 4,650          |
| Advertising                          |       | 6,827          | 3,678          |
| Bank Fees                            |       | 1,207          | 657            |
| Conferences & Trade Fairs            |       | 44,026         | 35,405         |
| Consulting                           |       | 166,009        | 160,301        |
| Depreciation                         |       | 10,169         | -              |
| Entertainment                        |       | 2,018          | 1,370          |
| Filing Fees                          |       | 310            | 290            |
| Foreign Currency Gains and Losses    |       | 2,240          | -              |
| Freight & Courier                    |       | 260            | 280            |
| General Expenses                     |       | 3,458          | 569            |
| Gift to Clients                      |       | 630            | 2,784          |
| Insurance                            |       | 22,811         | 14,118         |
| Interest Expense                     |       | 5,351          | 1,841          |
| Legal expenses                       |       | 5,075          | 6,350          |
| Memberships                          |       | 8,166          | 9,796          |
| Motor Vehicle                        |       | 123            | -              |
| Non-deductible expense               |       | 1,139          | 2,084          |
| Office Expenses                      |       | 1,957          | 2,670          |
| Printing & Stationery                |       | 75             | 672            |
| Subscriptions                        |       | 27,906         | 7,959          |
| Superannuation                       |       | 27,793         | 14,675         |
| Telephone & Internet                 |       | 2,906          | 3,294          |
| Travel and Accommodation             |       | 68,153         | 47,322         |
| Wages and Salaries                   |       | 252,664        | 139,762        |
| Website expenses                     |       | 7,061          | 14,197         |
| <b>Total Expenses</b>                |       | <b>672,836</b> | <b>474,723</b> |
| <b>Profit/(Loss) before Taxation</b> |       | <b>146,024</b> | <b>53,764</b>  |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

|                                        | NOTES | 2024   | 2023   |
|----------------------------------------|-------|--------|--------|
| <b>Income Tax Expense</b>              |       |        |        |
| Income Tax Expense                     |       | 66,233 | 28,400 |
| Total Income Tax Expense               |       | 66,233 | 28,400 |
| <b>Net Profit After Tax</b>            |       |        |        |
|                                        |       | 79,791 | 25,364 |
| <b>Net Profit After Dividends Paid</b> |       |        |        |
|                                        |       | 79,791 | 25,364 |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

# Balance Sheet

## Lithodat Pty Ltd As at 30 June 2024

|                                                     | NOTES | 30 JUNE 2024   | 30 JUNE 2023    |
|-----------------------------------------------------|-------|----------------|-----------------|
| <b>Assets</b>                                       |       |                |                 |
| <b>Current Assets</b>                               |       |                |                 |
| Cash on Hand                                        |       | 1,000          | 1,000           |
| <b>Bank Accounts</b>                                |       |                |                 |
| Amex Business Card                                  |       | -              | (30,950)        |
| American Express Platinum Busi                      |       | (46,938)       | -               |
| Business Telenet Saver                              |       | -              | 15,089          |
| CBA Business Account                                |       | 65,733         | 100             |
| CBA Business Online Saver                           |       | 10             | -               |
| CBA Capital Growth Account                          |       | 20,236         | -               |
| OLD BankWest Business Account                       |       | -              | 46,425          |
| Wise Euro                                           |       | 38             | -               |
| <b>Total Bank Accounts</b>                          |       | <b>39,078</b>  | <b>30,664</b>   |
| Accounts Receivable                                 |       | 125,400        | -               |
| <b>Total Current Assets</b>                         |       | <b>165,478</b> | <b>31,664</b>   |
| <b>Non-Current Assets</b>                           |       |                |                 |
| Trademarks                                          |       | 11,323         | 11,323          |
| <b>Property, Plant and Equipment</b>                |       |                |                 |
| Computer Equipment                                  |       | 15,805         | 5,636           |
| Less Accumulated Depreciation on Computer Equipment |       | (15,805)       | (5,636)         |
| <b>Total Property, Plant and Equipment</b>          |       | <b>-</b>       | <b>-</b>        |
| <b>Total Non-Current Assets</b>                     |       | <b>11,323</b>  | <b>11,323</b>   |
| <b>Total Assets</b>                                 |       | <b>176,800</b> | <b>42,986</b>   |
| <b>Liabilities</b>                                  |       |                |                 |
| <b>Current Liabilities</b>                          |       |                |                 |
| Child Support Payable                               |       | 1,099          | -               |
| GST                                                 |       | 19,159         | 6,030           |
| Income Tax Payable                                  |       | (17,887)       | (50,703)        |
| PAYG Withholdings Payable                           |       | 18,924         | 8,433           |
| Superannuation Payable                              |       | 8,812          | 2,323           |
| <b>Total Current Liabilities</b>                    |       | <b>30,106</b>  | <b>(33,917)</b> |
| <b>Non-Current Liabilities</b>                      |       |                |                 |
| Loan - Kristy Kohlmann                              |       | 10,000         | 20,000          |
| <b>Total Non-Current Liabilities</b>                |       | <b>10,000</b>  | <b>20,000</b>   |
| <b>Total Liabilities</b>                            |       | <b>40,106</b>  | <b>(13,917)</b> |
| <b>Net Assets</b>                                   |       | <b>136,694</b> | <b>56,903</b>   |
| <b>Equity</b>                                       |       |                |                 |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

Balance Sheet



|                   | NOTES | 30 JUNE 2024 | 30 JUNE 2023 |
|-------------------|-------|--------------|--------------|
| Retained Earnings |       | 135,694      | 55,903       |
| Share Capital     |       | 1,000        | 1,000        |
| Total Equity      |       | 136,694      | 56,903       |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

# Notes to the Financial Statements

## Lithodat Pty Ltd

### For the year ended 30 June 2024

#### 1. Statement of Significant Accounting Policies

The directors have determined that the company is not a reporting entity and accordingly, this financial report is a special purpose report prepared for the sole purpose of distributing a financial report to members and must not be used for any other purpose. The directors have determined that the accounting policies adopted are appropriate to meet the needs of the members.

The financial report has been prepared on an accrual basis and under the historical cost convention, except for certain assets, which, as noted, have been written down to fair value as a result of impairment. Unless otherwise stated, the accounting policies adopted are consistent with those of the prior year.

The accounting policies that have been adopted in the preparation of the statements are as follows:

##### Income Tax

The income tax expense for the year comprises current income tax expense. The company does not apply deferred tax. Current income tax expense charged to the profit or loss is the tax payable on taxable income calculated using applicable income tax rates enacted, or substantially enacted, as at 30 June 2024. Current tax liabilities are therefore measured at the amounts expected to be paid to the relevant taxation authority.

##### Property, Plant and Equipment

Property, plant and equipment is initially recorded at the cost of acquisition or fair value less, if applicable, any accumulated depreciation and impairment losses. Plant and equipment that has been contributed at no cost, or for nominal cost, is valued and recognised at the fair value of the asset at the date it is acquired. The plant and equipment is reviewed annually by directors to ensure that the carrying amount is not in excess of the recoverable amount from these assets. The recoverable amount is assessed on the basis of the expected net cash flows that will be received from the utilisation of the assets and the subsequent disposal. The expected net cash flows have been discounted to their present values in estimating recoverable amounts.

Freehold land and buildings are measured at their fair value, based on periodic, but at least triennial, valuations by independent external valuers, less subsequent depreciation for buildings.

Increases in the carrying amount of land and buildings arising on revaluation are credited in equity to a revaluation surplus. Decreases against previous increases of the same asset are charged against fair value reserves in equity. All other decreases are charged to profit or loss.

Any accumulated depreciation at the date of revaluation is offset against the gross carrying amount of the asset and the net amount is restated to the revalued amount of the asset.

##### Intangibles

Goodwill is recognised as the excess of the purchase price for a business acquired over the fair value of the net assets at the date of acquisition. Goodwill is assessed for impairment annually and is carried at cost less accumulated impairment losses.

##### Cash and Cash Equivalents

Cash and cash equivalents include cash on hand, deposits held on call with banks, other short-term highly liquid investments with original maturities of three months or less, and bank overdrafts.

##### Revenue Recognition

Revenue from the sale of goods is recognised upon the delivery of goods to customers.

Revenue from the rendering of services is recognised upon the delivery of the services to customers.

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These notes should be read in conjunction with the attached compilation report.

Revenue from commissions is recognised upon delivery of services to customers.  
Revenue from interest is recognised using the effective interest rate method.  
Revenue from dividends is recognised when the entity has a right to receive the dividend.

All revenue is stated net of the amount of goods and services tax (GST).

### **Goods and Services Tax**

Transactions are recognised net of GST, except where the amount of GST incurred is not recoverable from the Australian Taxation Office (ATO).

Receivables and payables are stated inclusive of the amount of GST receivable or payable. The net amount of GST recoverable from, or payable to, the ATO is included with other receivables or payables in the balance sheet.

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These notes should be read in conjunction with the attached compilation report.

## Appendix D

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Financial Statement FY2023-2024

# Annual Report

Lithodat Pty Ltd  
ABN 63 627 008 904  
For the year ended 30 June 2023

Prepared by Scott Partners Chartered Accountants

# Compilation Report

## Lithodat Pty Ltd For the year ended 30 June 2023

Compilation report to Lithodat Pty Ltd

We have compiled the accompanying special purpose financial statements of Lithodat Pty Ltd, which comprise the balance sheet as at 30 June 2023, the income statement, a summary of significant accounting policies and other explanatory notes. The specific purpose for which the special purpose financial statements have been prepared is set out in Note 1.

### The Responsibility of the Directors

The directors of Lithodat Pty Ltd are solely responsible for the information contained in the special purpose financial statements, the reliability, accuracy and completeness of the information and for the determination that the basis of accounting used is appropriate to meet their needs and for the purpose that financial statements were prepared.

### Our Responsibility

On the basis of information provided by the directors we have compiled the accompanying special purpose financial statements in accordance with the basis of accounting as described in Note 1 to the financial statements and APES 315 *Compilation of Financial Information*.

We have applied our expertise in accounting and financial reporting to compile these financial statements in accordance with the basis of accounting described in Note 1 to the financial statements. We have complied with the relevant ethical requirements of APES 110 *Code of Ethics for Professional Accountants*.

### Assurance Disclaimer

Since a compilation engagement is not an assurance engagement, we are not required to verify the reliability, accuracy or completeness of the information provided to us by management to compile these financial statements. Accordingly, we do not express an audit opinion or a review conclusion on these financial statements.

The special purpose financial statements were compiled exclusively for the benefit of the directors who are responsible for the reliability, accuracy and completeness of the information used to compile them. We do not accept responsibility for the contents of the special purpose financial statements.



Donald Shaw  
**Scott Partners Pty Ltd**  
Level 2, Suite 1, 2-6 Glenferrie Rd  
Malvern VIC 3144

05 October 2023 | 16:56 AEDT

Dated this ..... day of ..... 202...

# Directors Declaration

## Lithodat Pty Ltd

### For the year ended 30 June 2023

The directors have determined that the company is not a reporting entity and that this special purpose financial report should be prepared in accordance with the accounting policies outlined in Note 1 to the financial statements.

The directors of the company declare that:

1. The financial statements and notes, present fairly the company's financial position as at 30 June 2023 and its performance for the year ended on that date in accordance with the accounting policies described in Note 1 to the financial statements; and
2. In the directors' opinion there are reasonable grounds to believe that the company will be able to pay its debts as and when they become due and payable.

This declaration is made in accordance with a resolution of the Board of Directors.

Director: Fabian Kohlmann Fabian Kohlmann

Director: Gerd Moritz Walter Theile M. Theile

Director: Wayne Noble Wayne Noble

05 October 2023 | 04:40 EDT

Dated this ..... day of ..... 202...

# Income Statement

## Lithodat Pty Ltd

For the year ended 30 June 2023

|                                      | NOTES | 2023           | 2022           |
|--------------------------------------|-------|----------------|----------------|
| <b>Income</b>                        |       |                |                |
| Sales                                |       | 450,125        | 423,504        |
| <b>Total Income</b>                  |       | <b>450,125</b> | <b>423,504</b> |
| <b>Other Income</b>                  |       |                |                |
| Interest Income                      |       | 84             | 7              |
| Other Revenue                        |       | -              | 450            |
| Research & Development Offset Income |       | 78,278         | 90,907         |
| <b>Total Other Income</b>            |       | <b>78,362</b>  | <b>91,364</b>  |
| <b>Total Income</b>                  |       | <b>528,487</b> | <b>514,868</b> |
| <b>Expenses</b>                      |       |                |                |
| Accounting                           |       | 4,650          | 5,880          |
| Advertising                          |       | 3,678          | 3,738          |
| Bank Fees                            |       | 657            | 695            |
| Conferences & Trade Fairs            |       | 35,405         | 15,571         |
| Consulting                           |       | 160,301        | 246,377        |
| Depreciation                         |       | -              | 235            |
| Entertainment                        |       | 1,370          | 1,122          |
| Filing Fees                          |       | 290            | 276            |
| Foreign Currency Gains and Losses    |       | -              | 1,192          |
| Freight & Courier                    |       | 280            | 125            |
| General Expenses                     |       | 569            | 1,131          |
| Gift to Clients                      |       | 2,784          | 797            |
| Insurance                            |       | 14,118         | 7,884          |
| Interest Expense                     |       | 1,841          | -              |
| Legal expenses                       |       | 6,350          | 4,621          |
| Memberships                          |       | 9,796          | 4,355          |
| Non-deductible expense               |       | 2,084          | -              |
| Office Expenses                      |       | 2,670          | 161            |
| Printing & Stationery                |       | 672            | 950            |
| Subscriptions                        |       | 7,959          | 4,697          |
| Superannuation                       |       | 14,675         | 11,464         |
| Telephone & Internet                 |       | 3,294          | 2,238          |
| Travel and Accommodation             |       | 47,322         | 3,947          |
| Wages and Salaries                   |       | 139,762        | 114,876        |
| Website expenses                     |       | 14,197         | 4,896          |
| <b>Total Expenses</b>                |       | <b>474,723</b> | <b>437,227</b> |
| <b>Profit/(Loss) before Taxation</b> |       | <b>53,764</b>  | <b>77,641</b>  |
| <b>Income Tax Expense</b>            |       |                |                |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

|                                 | NOTES | 2023   | 2022   |
|---------------------------------|-------|--------|--------|
| Income Tax Expense              |       | 28,400 | 50,731 |
| Total Income Tax Expense        |       | 28,400 | 50,731 |
| Net Profit After Tax            |       | 25,364 | 26,910 |
| Net Profit After Dividends Paid |       | 25,364 | 26,910 |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

# Balance Sheet

## Lithodat Pty Ltd As at 30 June 2023

|                                                     | NOTES | 30 JUN 2023   | 30 JUN 2022   |
|-----------------------------------------------------|-------|---------------|---------------|
| <b>Assets</b>                                       |       |               |               |
| <b>Current Assets</b>                               |       |               |               |
| Cash on Hand                                        |       | 1,000         | 1,000         |
| <b>Bank Accounts</b>                                |       |               |               |
| Business Trans Acct                                 |       | 100           | -             |
| Amex Business Card                                  |       | (30,950)      | (6,354)       |
| Business Account                                    |       | 46,425        | 10,872        |
| Business Telenet Saver                              |       | 15,089        | 17,375        |
| <b>Total Bank Accounts</b>                          |       | <b>30,664</b> | <b>21,893</b> |
| Income Tax Refund                                   |       | 50,703        | 43,412        |
| <b>Total Current Assets</b>                         |       | <b>82,367</b> | <b>66,305</b> |
| <b>Non-Current Assets</b>                           |       |               |               |
| Trademarks                                          |       | 11,323        | 11,323        |
| <b>Property, Plant and Equipment</b>                |       |               |               |
| Computer Equipment                                  |       | 5,636         | 5,636         |
| Less Accumulated Depreciation on Computer Equipment |       | (5,636)       | (5,636)       |
| <b>Total Property, Plant and Equipment</b>          |       | <b>-</b>      | <b>-</b>      |
| <b>Total Non-Current Assets</b>                     |       | <b>11,323</b> | <b>11,323</b> |
| <b>Total Assets</b>                                 |       | <b>93,690</b> | <b>77,628</b> |
| <b>Liabilities</b>                                  |       |               |               |
| <b>Current Liabilities</b>                          |       |               |               |
| GST                                                 |       | 6,030         | 5,175         |
| PAYG Withholdings Payable                           |       | 8,433         | 10,779        |
| Superannuation Payable                              |       | 2,323         | 7,612         |
| Wages Payable - Payroll                             |       | -             | 5,418         |
| Sundry Creditor - ATO                               |       | -             | 809           |
| <b>Total Current Liabilities</b>                    |       | <b>16,787</b> | <b>29,792</b> |
| <b>Non-Current Liabilities</b>                      |       |               |               |
| Loan - Kristy Kohlmann                              |       | 20,000        | 16,296        |
| <b>Total Non-Current Liabilities</b>                |       | <b>20,000</b> | <b>16,296</b> |
| <b>Total Liabilities</b>                            |       | <b>36,787</b> | <b>46,088</b> |
| <b>Net Assets</b>                                   |       | <b>56,903</b> | <b>31,539</b> |
| <b>Equity</b>                                       |       |               |               |
| Retained Earnings                                   |       | 55,903        | 30,539        |
| Share Capital                                       |       | 1,000         | 1,000         |
| <b>Total Equity</b>                                 |       | <b>56,903</b> | <b>31,539</b> |

The accompanying notes form part of these financial statements. These statements should be read in conjunction with the attached compilation report.

# Notes to the Financial Statements

## Lithodat Pty Ltd

### For the year ended 30 June 2023

#### 1. Statement of Significant Accounting Policies

The directors have determined that the company is not a reporting entity and accordingly, this financial report is a special purpose report prepared for the sole purpose of distributing a financial report to members and must not be used for any other purpose. The directors have determined that the accounting policies adopted are appropriate to meet the needs of the members.

The financial report has been prepared on an accrual basis and under the historical cost convention, except for certain assets, which, as noted, have been written down to fair value as a result of impairment. Unless otherwise stated, the accounting policies adopted are consistent with those of the prior year.

The accounting policies that have been adopted in the preparation of the statements are as follows:

#### Income Tax

The income tax expense for the year comprises current income tax expense. The company does not apply deferred tax. Current income tax expense charged to the profit or loss is the tax payable on taxable income calculated using applicable income tax rates enacted, or substantially enacted, as at 30 June 2023. Current tax liabilities are therefore measured at the amounts expected to be paid to the relevant taxation authority.

#### Property, Plant and Equipment

Property, plant and equipment is initially recorded at the cost of acquisition or fair value less, if applicable, any accumulated depreciation and impairment losses. Plant and equipment that has been contributed at no cost, or for nominal cost, is valued and recognised at the fair value of the asset at the date it is acquired. The plant and equipment is reviewed annually by directors to ensure that the carrying amount is not in excess of the recoverable amount from these assets. The recoverable amount is assessed on the basis of the expected net cash flows that will be received from the utilisation of the assets and the subsequent disposal. The expected net cash flows have been discounted to their present values in estimating recoverable amounts.

Freehold land and buildings are measured at their fair value, based on periodic, but at least triennial, valuations by independent external valuers, less subsequent depreciation for buildings.

Increases in the carrying amount of land and buildings arising on revaluation are credited in equity to a revaluation surplus. Decreases against previous increases of the same asset are charged against fair value reserves in equity. All other decreases are charged to profit or loss.

Any accumulated depreciation at the date of revaluation is offset against the gross carrying amount of the asset and the net amount is restated to the revalued amount of the asset.

#### Intangibles

Goodwill is recognised as the excess of the purchase price for a business acquired over the fair value of the net assets at the date of acquisition. Goodwill is assessed for impairment annually and is carried at cost less accumulated impairment losses.

#### Cash and Cash Equivalents

Cash and cash equivalents include cash on hand, deposits held on call with banks, other short-term highly liquid investments with original maturities of three months or less, and bank overdrafts.

#### Revenue Recognition

Revenue from the sale of goods is recognised upon the delivery of goods to customers.

Revenue from the rendering of services is recognised upon the delivery of the services to customers.

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These notes should be read in conjunction with the attached compilation report.

Revenue from commissions is recognised upon delivery of services to customers.  
Revenue from interest is recognised using the effective interest rate method.  
Revenue from dividends is recognised when the entity has a right to receive the dividend.

All revenue is stated net of the amount of goods and services tax (GST).

### **Goods and Services Tax**

Transactions are recognised net of GST, except where the amount of GST incurred is not recoverable from the Australian Taxation Office (ATO).

Receivables and payables are stated inclusive of the amount of GST receivable or payable. The net amount of GST recoverable from, or payable to, the ATO is included with other receivables or payables in the balance sheet.

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These notes should be read in conjunction with the attached compilation report.

## Appendix E

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### Quality Management Policy



# **Quality Management System (QMS) Policy**

Digital-First Workplace Standards

Lithodat Pty Ltd  
ABN: 48 647 191 452  
December 2024

# Quality Management System (QMS)

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## Policy Overview

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Lithodat is committed to delivering scientifically accurate geoscience data platforms. Our **"Geology First"** approach ensures that data models and platform features are designed and validated by qualified geologists. This policy outlines our practical approach to data integrity, software testing, and scientific stewardship.

### 1. The "Geology First" Quality Principle

---

The core of our quality system is the integration of domain expertise into every stage of development. We ensure that:

- **Scientifically Sound Data Models:** Data structures are designed by geologists (not just developers) to ensure they reflect real-world geological relationships.
- **Expert Validation:** Our team of geologists manually and programmatically validates data to ensure scientific accuracy.
- **Geological AI:** Machine Learning workflows are reviewed by our AI team to ensure that automated outputs remain geologically plausible.

### 2. Practical Data Quality & QA/QC

---

We apply rigorous checks to ensure the data hosted on our platforms is of the highest possible quality:

- **Schema Validation:** Mandatory enforcement of controlled vocabularies (e.g., CGI/OGC standards, EarthBank and Isotopes.au ARDC ontologies) to prevent "dirty data" from entering the system.
- **Automated Logical Checks:** We use scripts to identify scientific inconsistencies (e.g., an analytical age that is geologically impossible given the stratigraphic context).
- **Provenance Tracking:** Every record includes metadata that traces the data back to its original laboratory source or publication, ensuring scientific accountability.

### 3. Software Testing & Development

---

To ensure platform reliability, we follow a consistent internal development process:

- **Peer Review:** No code is deployed to our platforms without being reviewed by a second developer to catch errors and security flaws.
- **Environment Separation:** We test all new features in a "Staging" environment before they are released to users (e.g., on EarthBank or Isotopes.au).
- **User Feedback Loops:** We work directly with academic partners to test tools in real-world scenarios before full deployment.

## 4. International Recognition & Certifications

---

Lithodat maintains external validation for our data stewardship through:

- **CoreTrustSeal (CTS) Certification:** EarthBank is officially certified as a Trusted Scientific Repository. This independently validates our long-term data stewardship and QA/QC systems.
- **World Data System (WDS) Membership:** As a Regular Member of the WDS (International Science Council), we comply with global standards for scientific data infrastructure.
- **AWS Infrastructure:** While Lithodat manages its own application security, we host our services on AWS to benefit from their world-class physical security and high-availability infrastructure.

## 5. FAIR Data Commitment

---

We ensure all data platforms managed by Lithodat follow the FAIR principles:

- **Findable:** Use of persistent identifiers (Litho-IDs) and rich geological metadata.
- **Accessible:** Data is available via standard web protocols with secure, role-based access.
- **Interoperable:** Use of standardized geological vocabularies and open formats (CSV, JSON).
- **Reusable:** Clear licensing (e.g., Creative Commons) and detailed laboratory provenance.

## 6. Continuous Improvement

---

Our continuous improvement process is practical and responsive. We:

- Conduct regular technical reviews after every major release.
- Actively solicit and implement technical feedback from our national research and government partners.
- Maintain an automated alerting system (AWS CloudWatch) to identify and fix performance issues immediately.

**Dr. Fabian Kohlmann**

*Chief Executive Officer*

December 2024

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**Lithodat Pty Ltd** | ABN: 48 647 191 452

Website: <https://lithodat.com>

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| 5 | Income Statement                  |
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| 8 | Notes to the Financial Statements |

## Appendix F

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### Health Safety and Environment Policy



# **Health, Safety & Environment Policy**

Digital-First Workplace Standards

Lithodat Pty Ltd  
ABN: 48 647 191 452  
December 2024

# Health, Safety & Environment Policy

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## Digital-First Workplace Standards

### Policy Overview

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Lithodat is committed to providing a safe work environment and operating in an environmentally responsible manner. As a compact, digital-first company, our HSE focus is on maintaining healthy remote-work habits, ensuring the security of our digital tools, and minimizing our environmental footprint through cloud-based operations.

### 1. Workplace Health & Safety (WHS)

---

Since Lithodat operates primarily as a remote and hybrid team, our health and safety focus is centered on the well-being of our staff in a digital environment:

- **Ergonomic Awareness:** We encourage all team members to maintain a safe and comfortable home-office setup, including proper chair support and monitor positioning to prevent strain.
- **Mental Well-being & Work-Life Balance:** We promote a culture of open communication. We manage fatigue by encouraging regular breaks and maintaining flexible working hours to prevent burnout.
- **Safe Digital Practices:** Safety at Lithodat includes digital safety. We ensure all team members use secure, encrypted connections and multi-factor authentication to protect both personal and company data.
- **Inclusivity:** We maintain a respectful workplace free from harassment. Any concerns can be raised directly with leadership for immediate and confidential resolution.

### 2. Environmental Sustainability

---

Lithodat minimizes its environmental impact by leveraging digital infrastructure rather than physical assets:

- **Cloud-First Infrastructure:** By hosting our platforms exclusively on AWS, we benefit from high-efficiency data centers and their commitment to renewable energy, reducing the need for local, energy-heavy servers.
- **Paperless Office:** We operate a near 100% paperless environment. All invoicing, contracts, and internal documentation are handled through digital signatures and cloud storage.
- **Reduced Travel:** We prioritize virtual meetings for internal and external collaboration, significantly reducing carbon emissions associated with commuting and business travel.
- **Equipment Responsibility:** We aim to maximize the lifespan of our IT hardware and ensure that end-of-life electronics are disposed of through local e-waste recycling programs.

### 3. Simple Incident Reporting

---

We maintain a straightforward process for reporting any health, safety, or environmental concerns:

- **Report:** Any issue (mental health strain, ergonomic injury, or digital security risk) should be reported directly to the CEO or COO.
- **Resolve:** Leadership will work directly with the affected individual to find a practical solution (e.g., adjusting workloads, providing equipment, or updating security protocols).
- **Prevent:** We review any incidents to see how our basic workflows can be improved to prevent a recurrence.

### 4. Summary of Commitment

---

At Lithodat we put our health, wellbeing and culture first for our people. We commit to the practical, daily safety of our people and the sustainability of our digital operations. We focus on being a **low-impact, high-wellbeing organization**.

**Dr. Fabian Kohlmann**

*Chief Executive Officer*

December 2024

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**Lithodat Pty Ltd** | ABN: 48 647 191 452

Website: <https://lithodat.com>

## Appendix G

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EarthBank



## EarthBank Project Summary

National Geoscience Data Platform for FAIR  
Geochemistry & Geochronology



GDAC-SA Advanced Analytics Platform RFQ  
Lithodat Pty Ltd  
ABN: 63 627 008 904

**Project Overview:** EarthBank is a \$21 million national initiative integrating over \$100 million in analytical instrumentation across 10 Australian university laboratories into a single FAIR-aligned digital platform. Co-developed by Lithodat Pty Ltd and AuScope Geochemistry Network, the platform demonstrates direct alignment with GDAC-SA requirements through proven capability in national-scale data standardization, multi-agency collaboration, and advanced geospatial analytics.

## 1. Overview

---

EarthBank (<https://ausgeochem.auscope.org.au>) is Australia's national cloud-based public repository for geochemistry, geochronology, isotopes, and thermochronology data. Co-developed by Lithodat Pty Ltd and AuScope Geochemistry Network, EarthBank represents a \$21 million investment in national research infrastructure funded by the Australian Research Council (ARC) and administered through AuScope Ltd.

The platform consolidates analytical data from over \$100 million in laboratory instrumentation across 10 Australian universities, making Australia's geoscience research outputs Findable, Accessible, Interoperable, and Reusable (FAIR). This directly mirrors GDAC-SA's mandate to centralize and standardize geoscience data at a national scale.

## 2. History & Funding

---

AuScope was established in 2006 as Australia's national geoscience research infrastructure provider through NCRIS, receiving over \$260 million in direct Commonwealth investment and catalyzing \$340 million total from all sources.

EarthBank evolved from AusGeochem (2019) to become a flagship \$21 million national initiative, integrating over \$100 million in analytical instrumentation from 10 university facilities into a unified FAIR-compliant digital platform. This substantial investment demonstrates Australia's recognition of national geoscience data infrastructure as critical research capability—the paradigm GDAC-SA seeks to implement for Saudi Arabia.

## 3. Key Collaborators

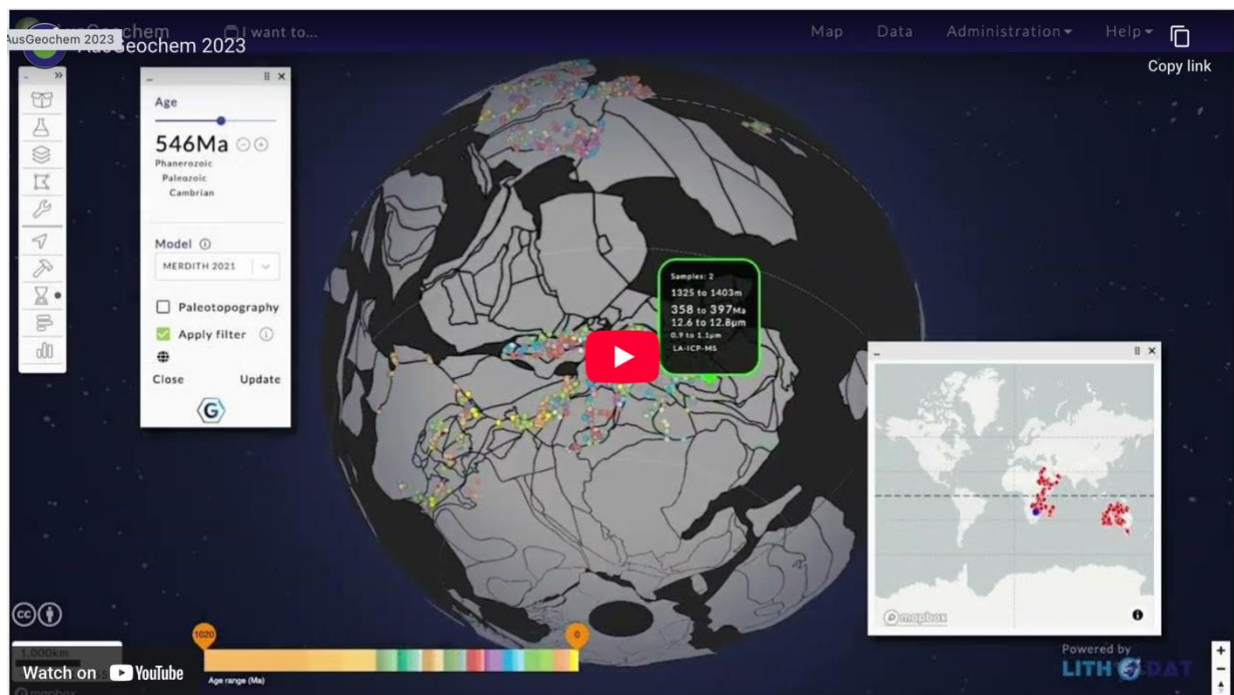
---

- **Universities:** Ten Australian laboratories (Curtin, Melbourne, ANU, UWA, Macquarie, Queensland, Adelaide, Monash, Tasmania, Wollongong) integrated into single FAIR-aligned platform.
- **Government:** Geoscience Australia and all state geological surveys collaborate on national isoscape mapping and geochemical baselines, demonstrating multi-agency capability applicable to GDAC-SA.
- **Industry:** 15% of users from industry (BHP, AngloAmerican, Chalice Mining), demonstrating translational value for mineral exploration.
- **Museums:** Museums Victoria collaboration digitized 43,500 historical specimens; 80,000+ additional specimens targeted, showcasing legacy data rescue capability relevant to GDAC-SA.
- **International:** OneGeochemistry initiative (global data standardization), GPlates community (tectonic-geochemical integration), EarthScope (USA), EPOS (Europe), IGSN e.V.
- **Leadership:** Professor Brent McInnes (Project Director, John de Laeter Centre, Curtin University).

## 4. Platform Capabilities

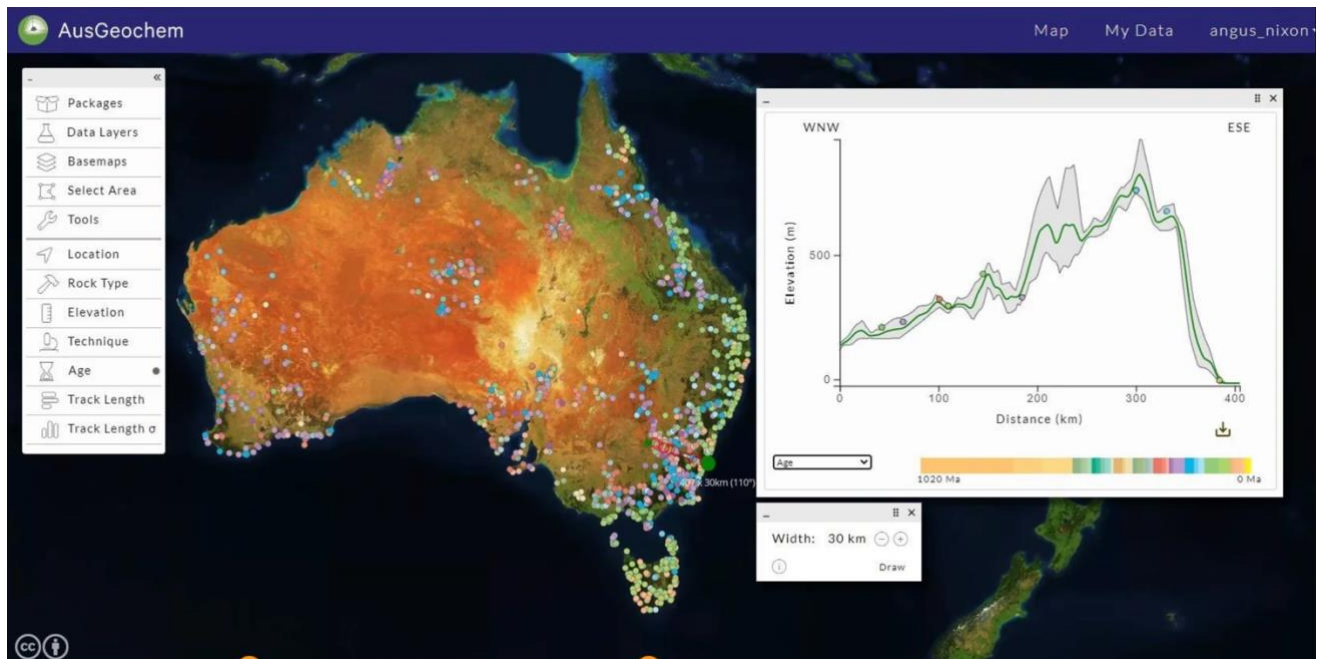
EarthBank supports six major analytical data types: U/Pb geochronology, Ar/Ar dating, fission-track thermochronology, (U-Th)/He thermochronology, major/trace element geochemistry, and thermal history modeling. EarthBank is powered by LithoSurfer – Lithodat's online software for interacting with geological data at scale. It provides scalable infrastructure for advanced geospatial analysis required by EarthBank and machine learning integration via REST API. Some examples of the features available in EarthBank include deep time reconstruction, swath profiles, chemistry dashboards, IDW interpolation, 3D visualization, DOI/IGSN minting, shapefile export, and comprehensive REST API for ML integration.

### 4.1 LithoPlates: Deep-Time Reconstruction



**Figure 1: LithoPlates enables deep-time paleogeographic reconstruction of geochemical data using 7 plate tectonic models to 1.8 Ga. This 4D capability (x, y, z, time) supports temporal analysis for mineral system formation through geological time—essential for GDAC-SA's advanced analytics requirements.**

## 4.2 Swath Profile Analysis



**Figure 2: Swath profile tool enables elevation and geochemical analysis along user-defined transects. This advanced spatial analytics capability supports mineral exploration targeting and tectonic interpretation—directly applicable to GDAC-SA's geospatial analysis requirements.**

## 4.3 Interactive Chemistry Dashboards



**Figure 3: Real-time geochemical dashboards with ternary plots (FeO-CaO-MgO oxides), TAS diagrams (Total Alkali-Silica plutonic classification), and spider diagrams (multi-element plots normalized to Chondrite 1995) enable rapid rock classification and geochemical interpretation—supporting GDAC-SA's advanced data visualization needs.**

## 5. FAIR Data & Impact

---

- **CoreTrustSeal Certification:** Highest international standard for trusted repositories, independently validating world-class data stewardship—demonstrating governance rigor required for national geological survey infrastructure.
- **FAIR Principles:** Findable, Accessible, Interoperable, Reusable data implementation enables research publication pipelines, national isotopic mapping, geochemical baselines, and museum digitization at scale.
- **Global Leadership:** OneGeochemistry initiative participation positions Australia as global leader in FAIR geoscience data infrastructure through internationally harmonized workflows.
- **Economic Impact:** \$15 ROI for every \$1 invested (Lateral Economics), \$912M mineral exploration value, \$458M gold discoveries, \$35M annual mining efficiencies.
- **Future Capabilities:** National isoscape mapping (Geoscience Australia partnership), GPLates integration for 4D mineral system analysis, expanded OneGeochemistry data sharing.

## 6. Publications & Recognition

---

**1. Boone, S.C., Dalton, H., Prent, A., Kohlmann, F., Theile, M., Noble, W., et al. (2022).** AusGeochem: An Open Platform for Geochemical Data Preservation, Dissemination and Synthesis. *Geostandards and Geoanalytical Research*, 46(2), 245-259. DOI: 10.1111/ggr.12419

- Highly cited establishing EarthBank as international reference for FAIR geochemical platforms. Demonstrates community-agreed schemas and collaborative development applicable to GDAC-SA's multi-stakeholder environment.

**2. Boone, S.C., Kohlmann, F., Noble, W., Theile, M., Beucher, R., Kohn, B., et al. (2023).** A geospatial platform for the tectonic interpretation of low-temperature thermochronology Big Data. *Scientific Reports*, 13, 8581. DOI: 10.1038/s41598-023-35776-3

- Demonstrates advanced analytical capabilities for thermochronology Big Data with GPLates integration for 4D reconstruction—showcasing complex temporal analyses essential for GDAC-SA advanced analytics.

For verification of project performance and technical details, contact:

Prof. Brent McInnes  
AuScope Earthbank Director  
b.mcinnnes@curtin.edu.au

# CLIENT TESTIMONIAL

## AuScope EarthBank National Geoscience Data Platform

### AuScope Ltd – Australian National Research Infrastructure

**Project:** EarthBank National Geoscience Data Platform

**Client:** AuScope Ltd

**Contract Value:** \$2.9M AUD

**Duration:** October 2019 – Present (6+ years, ongoing)

**Reference:** Prof. Brent McInnes, AuScope EarthBank Director

## Executive Summary

Lithodat has delivered exceptional outcomes for Australia's flagship national geochemical and isotope geoscience data platform - EarthBank. Over six years of continuous partnership, they have designed, built, and now operate a CoreTrustSeal-certified, FAIR-compliant repository integrating more than 12 million analytical results in geochemistry, geochronology, and isotope geoscience from federal and state geological surveys. EarthBank's additional accreditation by the International Science Council's World Data System confirms its international scientific standing and global interoperability. The platform is of demonstrably world-class quality and continues to exceed our expectations for technical excellence, reliability, and collaborative delivery.

## Performance & Results

### Technical Achievement

Lithodat successfully integrated four complex analytical domains, geochemistry, geochronology, isotopes, and thermochronology, into a unified, cloud-native platform hosted entirely on AWS. The sophistication of the unified data models, automated QA/QC pipelines, and spatial-temporal analytics positions EarthBank among the most advanced geological data repositories globally. EarthBank's certification as a CoreTrustSeal Trusted Digital Repository, together with accreditation by the International Science Council's World Data System, demonstrates both operational trustworthiness and international scientific standing, confirming alignment with the highest global standards for sustainable, high-quality research data infrastructure.

### Collaboration & Communication

Working with Lithodat has been seamless. Dr. Kohlmann and his team have demonstrated an exceptional ability to coordinate across academic institutions and laboratories, federal and state geological survey agencies, and national museums, harmonising diverse data standards and ingestion workflows. They communicate progress, risks, and opportunities proactively, translating complex technical decisions into clear, accessible information for a broad stakeholder community. Their responsiveness to evolving scientific and operational requirements has been consistently outstanding.

### Budget & Timeline Compliance

Over the six-year engagement, Lithodat has consistently delivered all milestones on time and within budget. The \$2.9M investment has yielded measurable returns through operational efficiency, research enablement, and international recognition of Australia's geoscience infrastructure. Their transparent cost management and commitment to value delivery have enabled multiple successful contract renewals.

### Speed & Agility

Despite the complexity of integrating 12+ million analytical records from diverse sources, Lithodat maintained rapid iteration cycles and responsive development. Their cloud-native architecture enables

continuous deployment of new features and data updates without service disruption. The team's ability to quickly adapt to new data types and analytical requirements has been critical to the platform's success.

## Key Metrics

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- **100% on-time delivery** of all contractual milestones (2019-2025)
- **12+ million analytical results** successfully integrated and quality-assured
- **Dual CoreTrustSeal and World Data System certification**, demonstrating adherence to the highest global standards for trusted scientific data repositories
- **FAIR compliance verified** – Findable, Accessible, Interoperable, Reusable
- **100% system uptime** SLA maintained across 6-year operational period
- **Multi-agency integration** – federal + state geological surveys harmonized
- **Zero critical defects** in production environment
- **API services operational** – supporting machine learning and advanced analytics

## Recommendation

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AuScope unreservedly recommends Lithodat for complex, multi-stakeholder national geoscience data infrastructure projects. Their technical expertise, scientific rigour, collaborative approach, and commitment to delivery make them an ideal partner for the GDAC-SA Advanced Analytics Platform. The six-year partnership on EarthBank demonstrates Lithodat's proven ability to deliver sustained value on mission-critical national research infrastructure.

### Signed



**Prof. Brent McInnes**

AuScope EarthBank Director

Email: b.mcinnnes@curtin.edu.au

Date: 19 December 2025

*Prof. McInnes is available to provide detailed verbal references regarding Lithodat's technical performance, collaborative capabilities, and suitability for the GDAC-SA project.*

*This testimonial reflects AuScope's assessment of Lithodat's performance on the EarthBank National Geoscience Data Platform project and is provided to support their bid for the GDAC-SA Advanced Analytics Platform tender.*

## Appendix H

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Isotopes.Au



## Isotopes.au Project Summary

Federal National Stable Isotope Platform – Multi-Agency Scientific Data  
Integration



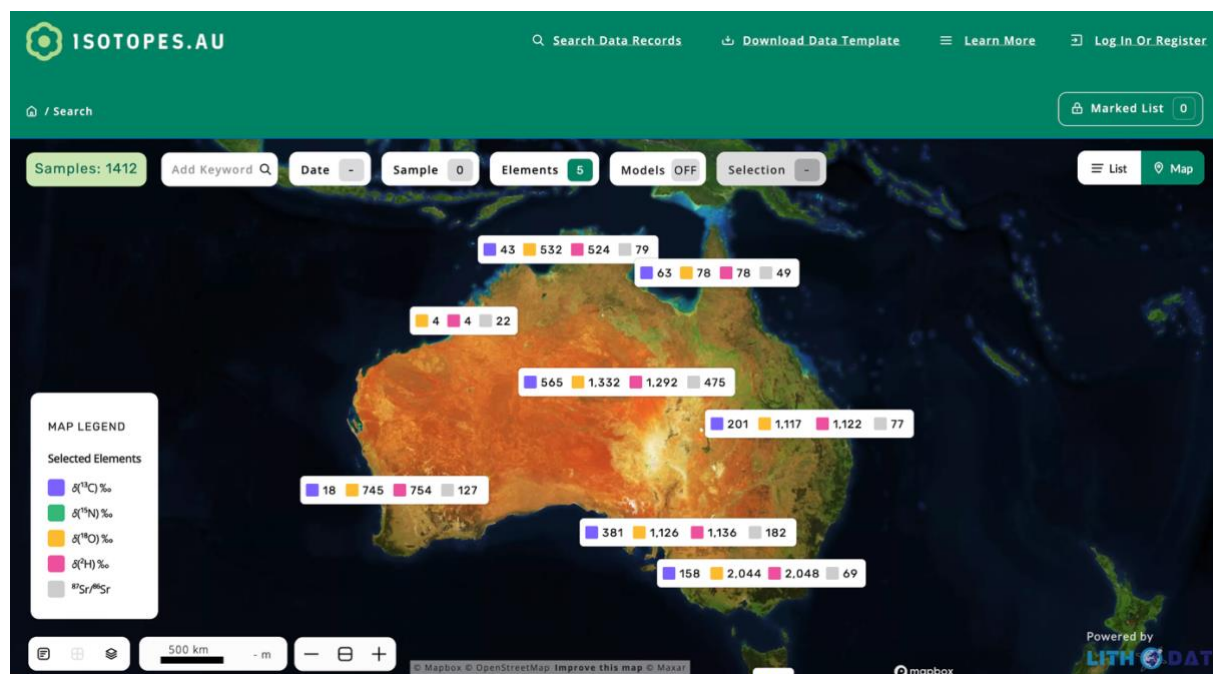
GDAC-SA Advanced Analytics Platform RFQ  
Lithodat Pty Ltd  
ABN: 63 627 008 904

**Project Overview:** is Australia's first national digital platform for standardizing, integrating, and querying stable isotope laboratory data across federal agencies and research institutions. Officially launched by **CSIRO** on **February 18, 2025**, the platform consolidates isotopic data—food's unique chemical fingerprints—into a single, open-access, trusted resource for verifying food provenance and sustainability claims.

## 1. Overview

Developed by **Lithodat Pty Ltd** in partnership with CSIRO, ANSTO, Geoscience Australia, NMI, and ARDC, Isotopes.au demonstrates advanced capability in multi-agency data harmonization, unified schema design, and FAIR-compliant scientific repository construction. The platform integrates six institutional data sources across three federal agencies, providing critical infrastructure for Australia's \$80 billion agriculture and food export industry.

**Platform Access:** Explore Isotopes.au at <https://app.isotopes.au>



**Figure 1: Isotopes.au platform interface showing interactive map-based data access and filtering capabilities for stable isotope measurements across Australia.**

## 2. History & Funding

**Contract Value:** \$400,000 AUD

**Duration:** May 2023 – December 2025 (ongoing hosting and support) | Program expansion through 2028

**Funding Sources:** Australian Research Data Commons (ARDC) + CSIRO co-funding

**Performance Rating:** Excellent (5/5) – 100% successful delivery

Lithodat delivered the complete platform on time and within budget, extending the LithoSurfer infrastructure to support isotope-specific data models. The project achieved all technical milestones, including integration of six institutional data sources, development of a 250+ field

ontology with 97% match rate, and implementation of hybrid automated/manual ingestion workflows.

The contract renewal for ongoing hosting and support through 2025, combined with program expansion to 2028, demonstrates exceptional client satisfaction. CSIRO Lead Scientist **Dr. Nina Welti** is available as a reference contact ([nina.welti@csiro.au](mailto:nina.welti@csiro.au)) to verify project performance and technical excellence.

### 3. Key Collaborators

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Isotopes.au represents a successful multi-agency federal collaboration across five major Australian government organizations:

**CSIRO** (Project Lead): Environmental isotopes laboratory data from two institutional sources (LabData and Data Access Portal), plus overall scientific direction and platform governance.

**ANSTO** (Australian Nuclear Science and Technology Organisation): Isoscape datasets and GNIP (Global Network of Isotopes in Precipitation) data, providing critical precipitation isotope measurements across Australia.

**Geoscience Australia**: National isotope surveys and hydrochemistry datasets from two institutional sources, contributing geological and hydrological isotope measurements.

**NMI** (National Measurement Institute): Isotope measurement standards and calibration data, ensuring measurement accuracy and consistency.

**ARDC** (Australian Research Data Commons): Infrastructure co-funding partner supporting national data infrastructure development.

*"Customers increasingly want to know where and how their food was sourced so they can make ethical and more sustainable choices. This is just the beginning of capturing Australia's wealth of isotopic data into one place—Isotopes.au—to help industries demonstrate how they're meeting environmental targets for greater transparency with trading partners and consumers."*

— Dr. Nina Welti, CSIRO Lead Scientist

### 4. Platform Capabilities

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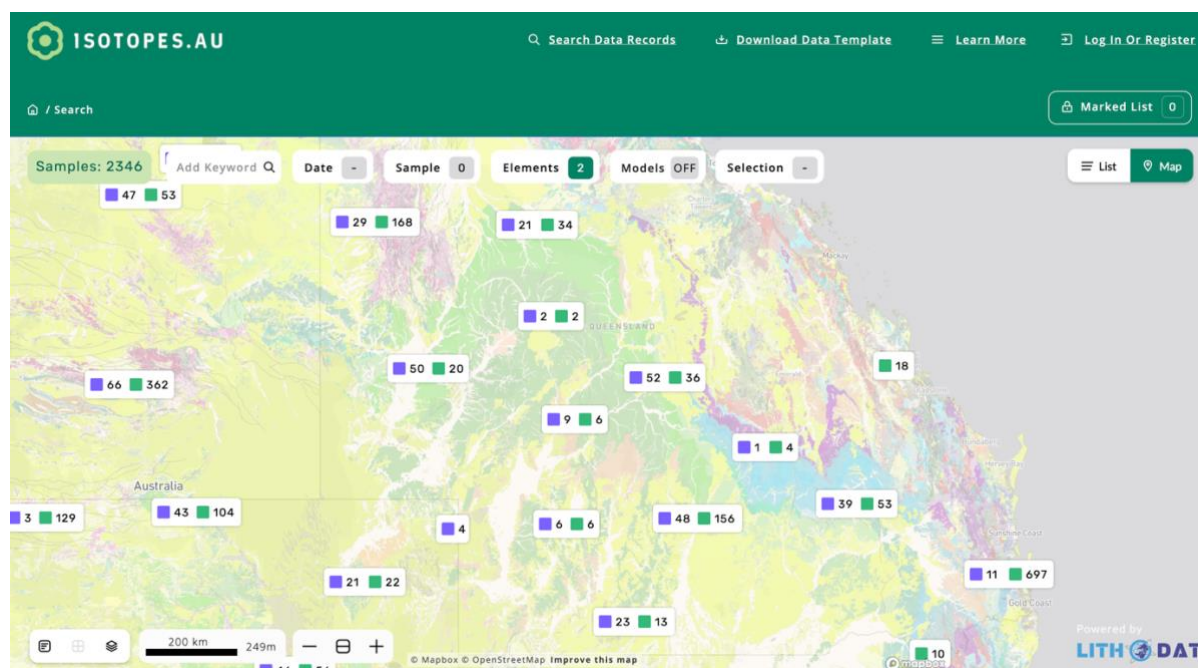
Isotopes.au integrates **six institutional data sources** through custom-developed adapters, harmonizing diverse data formats, measurement units, and metadata structures into a unified national data model. The platform achieved a **97% ontology match rate** across 250+ standardized field mappings after three iterative refinement rounds, demonstrating robust schema design.

**Supported isotope systems** include stable isotopes ( $\delta^{13}\text{C}$ ,  $\delta^{18}\text{O}$ ,  $\delta^2\text{H}$ ,  $\delta^{15}\text{N}$ ), radiogenic isotopes ( $^{87}\text{Sr}/^{86}\text{Sr}$ , Nd, Pb), and  $\delta^{34}\text{S}$  for mineral deposit studies. The platform handles complex isotopic measurements with full procedural metadata, quality control flags, and measurement uncertainties.

**Hybrid ingestion architecture** combines automated adapters for structured institutional data with a semi-automated import wizard for legacy and irregular datasets. The import wizard

achieved 93-96% field mapping accuracy with rejection rates under 6% after QA validation, completing most imports in under 15 minutes.

**API-first design** provides RESTful web services with comprehensive Swagger documentation, enabling programmatic access for analytics tools, external system integration, and automated workflows. The platform features interactive MapBox-powered geospatial visualization with GeoJSON rendering, filtering by isotope type, measurement method, material type, geographic region, and date range.



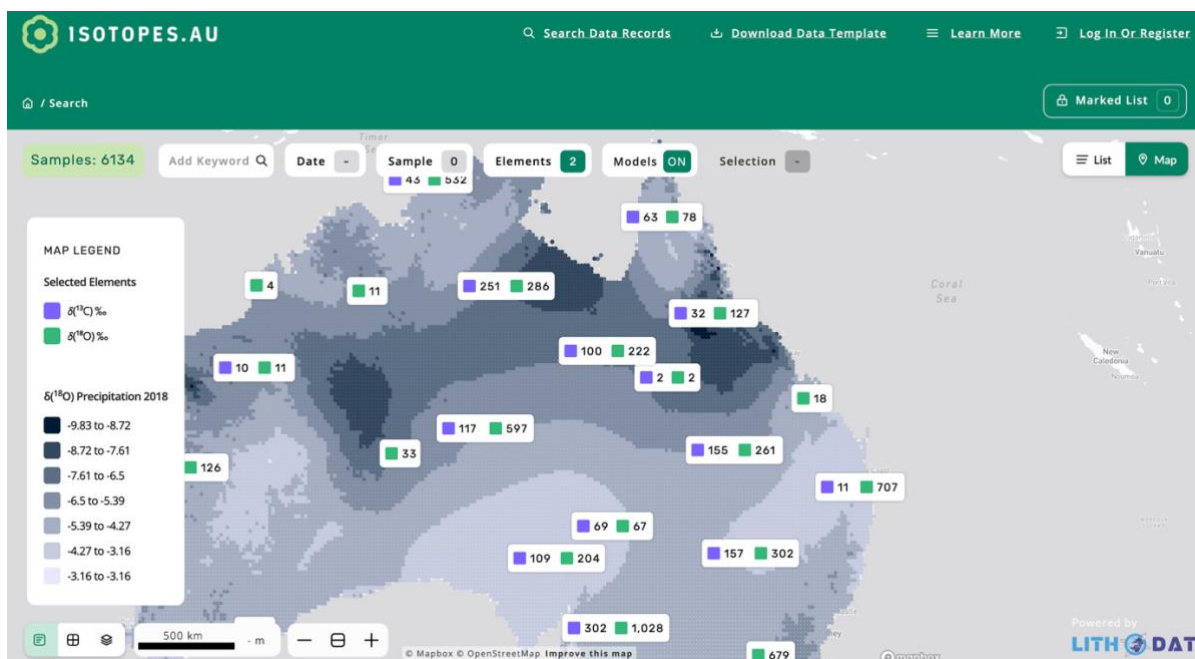
**Figure 2: Platform data visualization capabilities demonstrating geospatial mapping, isotope measurement display, and metadata filtering across multiple institutional data sources.**

## 5. FAIR Data & Impact

Isotopes.au implements comprehensive **FAIR data principles** (Findable, Accessible, Interoperable, Reusable) with persistent identifiers (DOI, IGSN), rich metadata using controlled vocabularies, open API access, and standardized data formats (JSON, GeoJSON, CSV). The LithoSurfer platform achieved **CoreTrustSeal certification** for the AusGeochem portal, validating long-term operational sustainability, robust security, documented governance, and organizational stability.

**National impact:** The platform supports Australia's **\$80 billion agriculture and food export industry** by enabling food provenance verification, sustainability claims validation, and trade regulation compliance. Isotopes.au aligns with the **National Agricultural Traceability Strategy** and provides critical infrastructure for consolidating data for trusted supply chains. Applications include regulatory reporting for environmental agencies, laboratory integration for live data upload, and commercial sector access for mining and environmental consulting.

**Future expansion:** The program extends through 2028 to broaden coverage beyond land-based measurements. Planned extensions include fisheries and aquaculture industry applications for tracking marine products through supply chains, plus food circularity support by underpinning safety standards for food reuse in production systems.



**Figure 3: Real-time analytics dashboard showing platform usage metrics, data ingestion statistics, and quality control monitoring across all institutional data sources.**

## 6. Publications

The Isotopes.au project has produced multiple peer-reviewed publications demonstrating scientific rigor and international recognition:

- Welti, N., Noble, W., Fraser, G., Flick, L., Gerber, C., Hawkins, S., et al. (2025). **Harmonizing Stable Isotope Data in Australia: The isotopes.au Platform for Enhanced Data Sharing and Collaboration.** EGU General Assembly 2025, Vienna. <https://doi.org/10.5194/egusphere-egu25-14534>
- Welti, N., Crawford, J., Fraser, G., Flick, L., Gerber, C., Hawkins, S., et al. (2024). **Federating Stable Isotope Data to Create a National Asset in Australia.** 13th International Conference on the Application of Stable Isotope Techniques in Ecological Studies, Fredericton, Canada.
- Flick, L., Crawford, J., Fraser, G., Gerber, C., Hawkins, S., Hughes, C., et al. (2024). **A Common Ontology for Stable Isotope Data.** 13th International Conference on the Application of Stable Isotope Techniques in Ecological Studies, Fredericton, Canada. <http://hdl.handle.net/102.100.100/636851?index=1>

**For verification of project performance and technical details, contact:**

Dr. Nina Welti  
CSIRO – Environmental Isotopes Laboratory  
Waite Campus  
Email: [nina.welti@csiro.au](mailto:nina.welti@csiro.au)

**Partners:** CSIRO, Geoscience Australia, ANSTO, NMI, ARDC

**Funding:** Australian Research Data Commons (ARDC) + CSIRO co-funding

# Appendix I

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## NRcan CATCH Database



## **NRCan CATCH Database**

Canadian Thermochronology National Data Platform



GDAC-SA Advanced Analytics Platform RFQ  
Lithodat Pty Ltd  
ABN: 63 627 008 904

**Project Overview:** The Canadian ThermoCHronology (CATCH) database is a \$400,000 CAD national platform integrating 51 years of thermochronology data. Developed by Lithodat for Natural Resources Canada's Geological Survey, CATCH demonstrates direct alignment with GDAC-SA requirements through proven capability in multi-decade data integration, legacy data digitization, and FAIR-compliant national database architecture delivered on-time and on-budget.

## 1. Overview

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The CATCH database is Canada's first unified national repository for low-temperature thermochronology data, integrating 1,427 samples from 91 publications spanning 1970-2021. Developed by Lithodat Pty Ltd (Fabian Kohlmann, Malcolm McMillan) with the Geological Survey of Canada (Dr. Jeremy Powell, Dr. Dawn Kellett), CATCH demonstrates national-scale data standardization directly applicable to GDAC-SA's multi-agency integration requirements.

**Performance:** \$400,000 CAD, 18 months (October 2021 - March 2023), delivered on-time and on-budget. Platform scale: 996 fission-track datapoints, 762 (U-Th)/He datapoints, 16,277 single grain measurements, published via DOI 10.4095/pdreqf905k and accessible through AusGeochem.

## 2. History & Funding

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Natural Resources Canada's Geological Survey (GSC) is Canada's national geoscience organization. Prior to CATCH, Canada had no central thermochronology database despite decades of published research—analogous to GDAC-SA's challenge of fragmented geological survey data.

In October 2021, GSC awarded Lithodat a \$400,000 CAD contract to design Canada's first national thermochronology data model, harvest all published research (1970-2021), and integrate with AusGeochem's FAIR platform. The project required systematic harmonization of legacy laboratory outputs from multiple provinces, automated QA/QC pipeline development, and extensive legacy data digitization from analogue paper reports and non-vectorized PDFs.

Deliverables included national data schema accommodating four thermochronometer types (apatite/zircon fission-track, apatite/zircon (U-Th)/He), standardization of 51 years of provincial data, complete harvesting from 91 publications, and integration with international reporting standards.

## 3. Key Collaborators

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**Client:** Natural Resources Canada - Geological Survey of Canada (equivalent to Saudi Geological Survey).

**Project Leadership:** Dr. Jeremy Powell (GSC project lead, first author), Dr. Dawn Kellett (GSC co-author), demonstrating government partnership excellence.

**Provincial Integration:** All Canadian provincial geological surveys contributed data requiring standardization across boundaries—demonstrating multi-agency harmonization applicable to GDAC-SA.

**Platform:** AusGeochem provides FAIR data access, enabling international research community access through open-source platform.

## 4. Platform Capabilities

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CATCH employs purpose-built data structure accommodating four thermochronometer types (Apatite/Zircon Fission Track, Apatite/Zircon (U-Th)/He) plus thermal history modeling. Canada's first unified national thermochronology schema handles complex kinetic parameters, accommodates legacy laboratory outputs, and provides ML-compatible, cloud-ready structure.



**Figure 1: CATCH database northern Canada compilation showing 1,427 sample locations with low-temperature thermochronology data (colored by geologic age: red = Precambrian, green = Paleozoic, blue = Mesozoic, yellow = Cenozoic). Demonstrates national-scale geospatial data integration with complete metadata capture for 51 years of published research. Data accessible via AusGeochem platform.**

**Advanced Features:** Single grain data preservation (16,277 measurements), fission-track length capture (27,848 measurements) from digitized histograms, automated QA/QC pipelines validated by GSC scientists, comprehensive provenance tracking, georeferencing workflow, and thermal history modeling integration.

## 5. FAIR Data & Project Performance

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**FAIR Principles:** Findable via DOI (10.4095/pdreqf905k) in GSC Open File series; Accessible through AusGeochem platform with no authentication barriers; Interoperable using community-agreed schemas; Reusable with complete metadata and ML-compatible structure.

**On-Time, On-Budget:** 18-month timeline met (October 2021 - March 2023), \$400,000 CAD managed with complete accountability, demonstrating reliable government partnership.

**National-Scale Achievement:** Standardized data across entire Canadian territory, harmonized provincial inconsistencies, integrated 51 years of data (1970-2021), harvested all published research from 91 sources—demonstrating systematic multi-decade integration applicable to GDAC-SA.

**Technical Excellence:** Automated data ingestion for multiple formats (digital, analogue paper, non-vectorized PDFs), QA/QC validation by GSC staff, and cloud-ready infrastructure supporting advanced analytics.

## 6. Publications & Recognition

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**Powell, J.W., Kellett, D.A., Kohlmann, F., and McMillan, M. (2025).** Canadian ThermoCHronology (CATCH) database: northern Canada compilation. *Geological Survey of Canada, Open File 9225*, 1 .zip file. DOI: [10.4095/pdreqf905k](https://doi.org/10.4095/pdreqf905k)

*Official government publication documenting Lithodat's design of Canada's first unified thermochronology data model and curation of 51 years of legacy data. Demonstrates multi-format extraction pipelines, automated QA/QC frameworks, and FAIR data implementation at national scale. The project's scope and methodologies are directly transferable to GDAC-SA's requirements for integrating historical geological survey data with modern analytics infrastructure.*

**For verification of project performance and technical details, contact:**

Dr. Jeremy Powell  
Geological Survey of Canada (Natural Resources Canada)  
Email: [jeremy.powell@nrcan-rncan.gc.ca](mailto:jeremy.powell@nrcan-rncan.gc.ca)

**Lithodat Pty Ltd** | ABN: 48 647 191 452  
Platform: <https://ausgeochem.auscope.org.au/>  
Support: [support@lithodat.com](mailto:support@lithodat.com)

# CLIENT TESTIMONIAL

## NRCan National Thermochronology & Thermal History Data Project

Natural Resources Canada – Geological Survey of Canada

**Project:** National Thermochronology & Thermal History Data Model Development

**Client:** Natural Resources Canada (Geological Survey of Canada)

**Contract Value:** \$400,000 AUD

**Duration:** October 2021 – March 2023 (18 months)

**Reference:** Dr. Jeremy Powell, Geological Survey of Canada (NRCan)

### Executive Summary

Lithodat delivered outstanding results on a transformational national data infrastructure project for the Geological Survey of Canada. In just 18 months, they designed and implemented a completely new national data model for thermochronology and thermal history datasets, standardized decades of provincial and historic data, and harvested all published thermochronology data across Canada. The project was completed on time, within budget, and has positioned Canada's geoscience community for modern machine learning and advanced analytics capabilities.

### Performance & Results

#### Technical Innovation & Quality

Lithodat's creation of Canada's first unified national thermochronology data model represents a significant scientific and technical achievement. The schema elegantly handles AHe, ZHe, AFT, and ZFT analytical methods while accommodating the complex kinetic parameters and metadata requirements of thermal history modeling. Their systematic approach to harmonizing inconsistent legacy laboratory outputs from multiple provincial jurisdictions demonstrates exceptional data engineering expertise and deep understanding of geoscience analytical workflows.

#### National Scope Delivery

The project's success in standardizing thermochronology data across Canada's entire territory-spanning provincial boundaries, historic laboratory variations, and decades of published research demonstrates Lithodat's capability to execute at true national scale. Their automated ingestion and QA/QC pipelines successfully processed diverse data formats while maintaining scientific integrity and provenance tracking. The resulting cloud-ready, ML-compatible structure positions Canadian researchers at the forefront of deep-time thermal analysis.

#### On-Time & On-Budget Execution

The 18-month delivery timeline from October 2021 to March 2023 was met without compromise to scope or quality. Lithodat's transparent project management, clear milestone planning, and proactive risk communication enabled Natural Resources Canada to maintain confidence throughout the engagement. The \$400,000 budget was managed with complete accountability, delivering exceptional value for a nation-building infrastructure investment.

#### Speed & Efficiency

Despite the complexity of designing new national data models, harvesting historic published data, and building automated QA/QC systems, Lithodat maintained impressive velocity throughout the project. Their ability to rapidly prototype schema designs, gather stakeholder feedback, and iterate toward consensus

accelerated delivery without sacrificing quality. The efficient integration with their LithoSurfer platform architecture eliminated redundant development effort.

## Key Metrics

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- **18-month on-time delivery** – completed March 2023 as scheduled
- **National data standardization** – complete coverage across all Canadian provinces
- **All historic datasets harmonized** – decades of published research integrated
- **Automated QA/QC pipelines operational** – validated by GSC scientific staff
- **Continued partnership engagement** – ongoing relationship demonstrates satisfaction
- **ML-compatible structure delivered** – cloud-ready for advanced analytics
- **Complete data harvesting** – all published thermochronology data across Canada
- **LithoSurfer platform compatibility** – seamless ecosystem integration

## Recommendation

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Natural Resources Canada strongly recommends Lithodat for national-scale geoscience data infrastructure modernization projects. Their demonstrated ability to design innovative data models, standardize diverse provincial datasets, and deliver complex technical solutions on time and within budget makes them an ideal partner for the GDAC-SA Advanced Analytics Platform. The NRCan thermochronology project proves their capability to transform fragmented legacy data into unified, analytics-ready national resources.

## Reference Contact



**Dr. Jeremy Powell**

Geological Survey of Canada (NRCan)

Email: [jeremy.powell@nrcan-rncan.gc.ca](mailto:jeremy.powell@nrcan-rncan.gc.ca)

Date: 18-12-25

*Dr. Powell is available to provide detailed verbal references regarding Lithodat's technical performance, national-scale delivery capabilities, and suitability for the GDAC-SA project.*

*This testimonial reflects Natural Resources Canada's assessment of Lithodat's performance on the National Thermochronology & Thermal History Data Model Development project and is provided to support their bid for the GDAC-SA Advanced Analytics Platform tender.*

## Appendix J

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### Publication Summary



## **Lithodat Team Academic References**

### **Bibliography**

GDAC-SA Advanced Analytics Platform RFQ  
Lithodat Pty Ltd  
ABN: 63 627 008 904

## 1. Geochemistry Data Platforms (13 references)

*Cloud-based platforms for geochemical data management, preservation, and analysis*

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| Nixon AL, Boone SC, Gréau Y, Kohlmann F, Theile M, Noble W, Prent A, Foden J, Barnes S, Wainwright AN, Wade B, Goemann K, Gilbert S, Evans NJ, Dalton H, Ware B, McInnes BIA | Volcanoes to vugs: Demonstrating a FAIR geochemistry framework with a diverse application of major and trace element data through the AuScope EarthBank platform | Chemical Geology 696                                  | 2025 | <a href="https://doi.org/10.1016/j.chemgeo.2024.122494">10.1016/j.chemgeo.2024.122494</a> |
| Boone SC, Dalton H, Prent A, Kohlmann F, Theile M, et al.                                                                                                                    | AusGeochem: An open platform for geochemical data preservation, dissemination and synthesis                                                                      | Geostandards and Geoanalytical Research 46(2):245-259 | 2022 | <a href="https://doi.org/10.1111/ggr.12419">10.1111/ggr.12419</a>                         |
| Klöcking M, Wyborn L, Lehnert KA, Ware B, Prent AM, Profeta L, Kohlmann F, et al.                                                                                            | Community recommendations for geochemical data, services and analytical capabilities in the 21st century                                                         | Geochimica et Cosmochimica Acta 351:192-205           | 2023 | <a href="https://doi.org/10.1016/j.gca.2023.04.024">10.1016/j.gca.2023.04.024</a>         |
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| Kohlmann F, Noble W, Qin X, Highton J, et al.                                                                                                                                | Deep-Time Digital Twins: Integrating LithoPlates with the EarthBank Platform                                                                                     | EGU General Assembly 2025                             | 2025 | <a href="#">EGU25</a>                                                                     |
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| Noble W, Kohlmann F, Prent AM, Boone S, et al.                                                                                                                               | Opening the full Potential of Geochemical Big Data utilising AusGeochem                                                                                          | Goldschmidt 2022                                      | 2022 | <a href="#">Goldschmidt</a>                                                               |
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| Theile M, Noble W, Beucher R, et al.                                                                                                                                         | From Fieldwork to Publication with AusGeochem: An Open end-to-end Solution for managing FAIR Sample Based Geoscience Data                                        | EGU General Assembly 2023                             | 2023 | <a href="#">EGU23</a>                                                                     |
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*Low-temperature thermochronology, fission track analysis, and geochronological studies*

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*AI and machine learning applications in mineral prospectivity and exploration*

| Authors                                                                                                                                                         | Title                                                                                                                                                                           | Source                        | Year | DOI / Link                                                                              |
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#### 4. Fractal & Multifractal Geochemistry (10 references)

*Fractal and multifractal modeling for geochemical anomaly detection and mineral exploration*

| Authors                                                         | Title                                                                                                                                                                                  | Source                                   | Year | DOI / Link                                                                                    |
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## 6. Critical Minerals & Rare Earths (4 references)

*Research on critical minerals, rare earth elements, and strategic resources*

| Authors                                                                  | Title                                                                                                                                                                          | Source                              | Year | DOI / Link                                                                                    |
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*Petroleum systems analysis, source rock characterization, and basin modeling*

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*Remote sensing and GIS methods for hydrothermal alteration mapping and mineral prospectivity*

| Authors                                                               | Title                                                                                                                                                                                           | Source                                | Year | DOI / Link                                                                                      |
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## 10. Volcanology (2 references)

*Volcanic systems, hazard assessment, and magmatic processes*

| Authors                         | Title                                                                                                                            | Source                                            | Year | DOI / Link                                                                                      |
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## 11. Environmental & Sustainability (2 references)

*Environmental monitoring, contamination reporting, and sustainability in geosciences*

| Authors                                        | Title                                                                                                                                             | Source                  | Year | DOI / Link                                                                  |
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| Dimech K                                       | 'PROJECT PATHWAY' TO NET ZERO: SCOPE 3 CARBON IN WATER CAPITAL PROJECTS FROM TENDER TO AS BUILT                                                   | Ozwater 2025 Conference | 2025 | <a href="#">Ozwater25</a>                                                   |

## 12. Geological Survey & Exploration (5 references)

*Regional geological surveys, stream sediment analysis, and exploration geochemistry*

| Authors                                                       | Title                                                                           | Source                             | Year | DOI / Link                                                                              |
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### 13. Petroleum Exploration Technology (4 references)

*Advanced technologies and methods for petroleum exploration*

| Authors                                                      | Title                                                                                                                                   | Source                      | Year | DOI / Link                                  |
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| Mezghani MM, Najjar NF, AbuAli M, Zuhlke R, Inan S, Allen CK | Iterative and repeatable workflow for comprehensive data and processes integration for petroleum exploration and production assessments | US Patent 10,713,398        | 2020 | <a href="#">US10713398B2</a>                |
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| Klusman RW, Saeed MA, Abu-Ali MA                             | The potential use of biogeochemistry in the detection of petroleum microseepage                                                         | AAPG Bulletin 76(6):851-863 | 1992 | <a href="#">AAPG</a>                        |
| Zhang SH, Xu Y, Abu-Ali M                                    | Characterizing stratigraphic traps using improved waveform classification seismic facies analysis: central Saudi Arabia                 | First Break 34(12)          | 2016 | <a href="#">10.3997/1365-2397.fb2016098</a> |

## Appendix K

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### Curriculum Vitae

# Dr Fabian Kohlmann

Curriculum Vitae



## Dr Fabian Kohlmann

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## Present Employment

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### Jan 2019 – Present | Managing Director

#### Lithodat Pty. Ltd.

Melbourne, Australia

- Managing Director of cloud-hosted exploration data & software company.
- Responsible for day-to-day management, direction, marketing, acquisition and operation. Also providing subject matter expertise to the Lithodat team supporting product development and data mining efforts and providing training courses to clients in the resource industry.

### Jun 2025 – Present | Honorary Principle Research Fellow

#### The University of Melbourne

Melbourne, Australia

- Supervision and mentoring of students in EarthSciences in the MEGs group.
  - Working closely with Prof. Andrew Gleadow, Dr. Brandon Mahan and Dr. Malcolm McMillan.
  - Teaching every year a one week graduate level VIEPS course Spatial Data Analytics and Geoprocessing EARTH90060 course.
- 

## Previous Employment

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### Aug 2019 – Jul 2022 | Adjunct Lecturer

#### The University of Adelaide

Adelaide, Australia

- Assisting Adelaide Earth Science students with data science in geology.
- Preparing lecture for graduate level for advances GIS workflows and data base use in research.

### Jul 2019 – May 2025 | Honorary Research Fellow

#### The University of Melbourne

Melbourne, Australia

- Supervision and mentoring of students in EarthSciences in Data Science in Geology.
- Working closely with Prof. Andrew Gleadow, Dr. Brandon Mahan and Prof. B. Kohn.

## **Aug 2016 – Dec 2019 | Director**

### **Pangeo Exploration Limited**

Didcot, UK

- Director position in a six men consulting company aimed at the petroleum and mining industry.
- My role within this consulting company was the management and provision of expertise knowledge and solutions for thermochronology, thermal history modelling, GIS optimization, PaleoDEM creations, landscape evolution and Paleoclimate modelling.

## **Jun 2014 – Nov 2018 | Senior Geodynamics Geoscientist**

### **Halliburton, Landmark – Neflex Exploration Insights**

Abingdon, UK

- In charge of managing the development of Global Paleo Digital Elevation Models from 595Ma to present, currently version 2.
- Geoprocessing of exploration data using advanced ArcGIS techniques.
- Paleoreconstructions in PaleoGIS using the Landmark-Neflex Plate Model and support of staff how to use it.
- Mentoring and training of staff in the correct use of thermochronology in exploration, thermal history modelling and advanced geoprocessing in GIS.
- Delivering training courses in geochronology and thermochronology within Halliburton.
- Support and R&D of our Paleoclimate modelling and the Earth System Model.
- Managing the release, delivery and support of all PaleoDEM products to our clients.
- Supervision of publishing all PaleoDEM documents and data in GIS and on the webmapper.
- Set-up and management of the in-house Paleo-elevation database and the thermochronology database.
- Improving workflows and geoprocessing for Geodynamics, Earth System Modelling and GDE mapping teams.
- Modelling sequence stratigraphic effects into PaleoDEMs, such as continental slope evolution, MFS and LST scenarios, increased clastic inputs and many more.

## **May 2013 – Jun 2014 | Geodynamics Geoscientist**

### **Neflex Petroleum Consultants Ltd**

Abingdon, UK

- Developing the first version of Neflex global PaleoDEMs from 595 Ma to present.
- Geoprocessing of geodynamic data in ArcGIS.
- Paleoreconstructions of exploration data in PaleoGIS using the Neflex Plate Model.
- Support of the Geodynamics team, generating and fixing geodynamic overlays.
- QA/QC of plate model and Gross Depositional Environment (GDE) maps, during PaleoDEM generation.

## **Oct 2010 – Apr 2013 | Postdoctoral Research Fellow in Geodynamics / Thermochronology**

### **Department of Earth Sciences**

University of Bergen, Norway

- Continues research in structural geology and low-temperature thermochronology.
- Employed through the Statoil funded "Earth System Modeling (ESM)" project with UiB.
- Main research was focused on the thermodynamical evolution of the Norwegian margin in the Hardangerfjord and Sognefjord areas, applying fission track analysis, (U-Th)/He dating and inverse thermal history modeling.
- Comparing the offshore sedimentary record with the onshore denudation history.
- Modelling landscape evolution and erosion histories derived from thermochronological data for Source-to-sink studies.

- Responsible for the fission track lab at UiB, including maintenance and introduction of new students and guest researchers in the fission track methodology.
- Established contacts with the Physics Department at UiB to continue my research in basic fission track formation processes at the nanoscale.
- Other low-temperature thermochronological study areas are located in northern Norway, Southern Argentina and Northern Canada.
- Successful supervision of 5 MSc students and co-supervision of 2 PhD students.
- Lecturing thermochronology as part of the applied Geochronology course.

## **Apr 2010 – Oct 2010 | Researcher in Geodynamics/Thermochronology**

### **Department of Earth Sciences**

University of Bergen, Norway

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## **Education**

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### **Feb 2005 – Oct 2010 | PhD in Earth Sciences**

#### **School of Earth Sciences**

The University of Melbourne, Victoria, Australia

### **Nov 1999 - June 2003 | Diplom (equivalent to MSc) in Geology/Palaeontology**

**Grade:** High Distinction (1.3 = 94%)

#### **Department for Geo- and Life Sciences**

Ludwig-Maximilians-Universitaet (LMU), Munich, Germany

# Dr. Wayne Noble

Chief Information Officer



**GDAC-SA Tender Role:** Technical Director | 12+ Years Technical Leadership | Scientific Software Developer | Enterprise-Scale Agile Transformation

**Technical Director - Agile Leadership & Organizational Change**

## Current Position

### Technical Director

**Lithodat Pty Ltd | Melbourne, Australia** 2019 - Present (6+ years)

- Technical strategy and platform architecture leadership
- Oversee all technology initiatives and development teams
- Guide technical decision-making and standards
- Lead technical aspects of GDAC-SA tender response
- Mentor and develop technical talent
- Drive innovation in geoscience data management

## Core Expertise

- Agile Transformation & Coaching
- Enterprise Software Architecture
- Team Leadership & Development
- Organizational Change Management
- Software Development Lifecycle
- Product Development & Strategy
- DevOps & Continuous Delivery
- Technical Consulting

## Key Professional Experience

### MYOB - 13 Years 7 Months Total Experience

#### Agile Coach / Delivery Coach

**MYOB | Melbourne, Australia** October 2018 - October 2019

- Coached teams and leaders in agile practices and mindset
- Facilitated organizational transformation initiatives
- Developed and delivered training programs
- Helped teams improve velocity and delivery quality
- Supported leadership in adopting agile methodologies

#### Agile Coach

**MYOB | Melbourne, Australia** October 2011 - December 2015

- Managed engineering teams delivering enterprise software
- Implemented DevOps practices and continuous delivery
- Drove technical excellence and engineering standards
- Created a model for continuous improvement in teams

#### Developer / Senior Developer

**MYOB | Melbourne, Australia** May 2003 - October 2011

- Architected key product features and modules
- Mentored junior and mid-level developers
- Established coding standards and best practices
- Drove adoption of automated testing

## Consulting & Advisory Roles

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### Agile Transformation Consultant

**MYER | Melbourne, Australia** *November 2019 - April 2020*

- Guided retail technology transformation
- Coached leadership and teams in agile practices
- Facilitated strategic planning sessions
- Helped establish agile centers of excellence

### Agile Transformation Consultant

**CPA Australia | Melbourne, Australia** *February 2018 - October 2018*

- Advised on digital transformation strategy
- Provided guidance on team structure and practices

### Agile Transformation Consultant

**Australia Post | Melbourne, Australia** *October 2017 - February 2018*

- Coached multiple teams across the organization
- Supported leadership in change management

### Agile Transformation Consultant

**Colonial First State | Melbourne, Australia** *March 2017 - July 2017*

- Facilitated quality uplift through adoption of unit testing
- Recommended improvements to delivery practices

### Agile Transformation Consultant

**Yarra Valley Water | Melbourne, Australia** *July 2016 - March 2017*

- Established a tribe cadence of planning
- Coached people new to the role of Product Owners
- Recommended improvements to delivery practices

### Agile Transformation Consultant

**Hansen Technologies | Melbourne, Australia** *January 2016 - July 2016*

- Introduced Agile working
- Trained team members in Scrum practices and rhythms
- Coached the wider business in the changes required to adopt Agile

## Education

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### Graduate Diploma in Software Development (with Distinction)

**RMIT University | Melbourne, Australia** *2002 - 2006*

- Completed with distinction

### PhD - Earth Science

**Latrobe University | Melbourne, Australia** *1992 - 1997*

- Thermochronology study across eastern Africa
- Highly developed analysis and analytic skills

### Bachelor of Science (Honours)

**La Trobe University | Melbourne, Australia** *1986 - 1990*

- First Class Honours
- Strong foundation in geology and research

## Technical Skills

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- **Leadership:** Agile coaching, team building, organizational change, strategic planning
- **Software Development:** Full-stack development, architecture design, DevOps practices
- **Methodologies:** Agile, Scrum, Kanban, SAFe, DevOps, Lean
- **Technical:** C#, Java, Python, SQL, cloud platforms (AWS, Azure), CI/CD pipelines
- **Domain Expertise:** Enterprise software, scientific computing, data platforms, Earth Science, thermochronology
- **Tools:** JIRA, Confluence, Git, Jenkins, Docker, Kubernetes

## Publications & Speaking

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- Multiple conference presentations on agile transformation
- Guest lectures at University of Melbourne on software engineering
- Published articles on technical leadership and team dynamics
- Active contributor to agile and DevOps communities

# Keith Dimech

Chief Operating Officer



**GDAC-SA Tender Role:** Chief Operating Officer | Database Development Expert | 2024 Victorian Infrastructure Award Winner | Industry Leadership (AWA, Beyond Zero)

**Chief Operating Officer - Environment and Sustainability Manager**

## Current Positions

### Chief Operating Officer (COO)

**Lithodat Pty Ltd | Melbourne, Australia** 2019 - Present (6+ years)

- Oversee all operational aspects of the company
- Lead GDAC-SA tender preparation and submission
- Manage client relationships and project delivery
- Database development and data management strategy
- Business development and strategic partnerships
- Financial planning and operational efficiency

### Environment and Sustainability Manager

**John Holland Water | Melbourne, Australia** September 2021 - Present (3+ years)

- Developed and rolled out the Project Pathway to Net Zero Sustainability strategy for John Holland and Melbourne Water, now utilised across multiple projects and contractors
- Environmental and Sustainability Manager for Hobsons Bay Main and Western Treatment Plant RRRC projects, with a combined project value of \$900 million
- Winner of the 2024 Victorian Infrastructure Project Innovation Award - Australian Water Association
- Embedded in the corporate Environment and Sustainability team, undertaking policy writing, tendering, auditing and compliance
- Authorship of the ISO 9001 and 14001 accredited - Environmental, Sustainability and Safety Management Plan / Policy / Procedures
- Member of the HSEQ leadership team, including times as acting Safety Manager
- Ensuring the project achieves and abides by all approvals from various regulators required
- Delivered Australia's First Carbon Neutral Project - led the team to ensure lowest possible carbon emissions onsite through use of new technologies, sustainability through design and detailed materials tracking

## Major Awards & Recognition

### 2024 Victorian Infrastructure Project Innovation Award

**Australian Water Association (AWA)**

- Recognized for innovative approaches to environmental data management on infrastructure projects
- Implemented novel database systems for tracking environmental compliance

### 2025 Australian Infrastructure Project Innovation Award

**Australian Water Association (Finalist - Announced May 2025)**

### 2025 Built Environment and Infrastructure

**Banksia Award (Finalist - Announced April 2025)**

### 2024 John Holland Project of the Year

### 2022 'Significant Personal Contribution'

**Melbourne Water Award**

## Core Expertise

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### Sustainability Management

- Innovator in the management of digital construction material tracking and carbon accounting, developing a cloud-based SQL database to track material quantities and embodied carbon through project lifecycle
- Proficient in carbon accounting, social expenditure data collection, and regulatory reporting, including SBTi, Building Equality Policy, Indigenous Procurement Policy, and National Greenhouse Energy Reporting Scheme
- Founder of the Beyond Zero – Australian Water Association National Sustainability Specialist Network
- Deep knowledge of the Australian and International Carbon Credit Markets
- Embedding carbon into the cost planning, tendering, and quantity surveying decision-making procedures
- Prior member of the Low Carbon in Concrete Working Group with MTIA; strong understanding of new and emerging low-carbon construction materials, practices, and methods

### Environmental Management

- Author of, and ensure compliance with, various environmental management plans, environmental permits, and procedures for cultural heritage, biodiversity, groundwater, contaminated land, noise, air quality and waste
- Extensive experience in approvals management, including EPAV, Council (Native Vegetation, Planning, Heritage Victoria, Traditional Owner Corporations), Parks Vic/DEECA, and Utilities
- Founder of an industry working group, applying for a CRC-P Grant and build a federal relational and geospatial database platform to store, analyse and report on federal environmental contamination and biodiversity data
- Deep knowledge in geotechnical field preparation and execution with a focus on soil and groundwater data collection; expert knowledge in contaminated land field works

### Business Corporate Roles

- Corporate strategy, policy and procedure writing. Lead auditor and compliance manager across multiple projects
- Worked on, helped price large Energy, Water, Road and Rail Tenders, including successful bids for Northeast Link, Western Trunk Sewer, Resource Recovery & Reuse Complex, Suburban Rail Loop and Belmont Desalination Plant. Awaiting results on Tarraleah Hydroelectric upgrade, Aurora Treatment Plant and Healesville Treatment Plant
- Management, forecasting and budgeting of multiple million-dollar packages, including environment and sustainability expenses, construction and operation of water treatment plants and spoil cartage and disposal
- Tender, scope, and manage dozens of subcontracts valued between \$10k - \$2m
- 4 Direct reports and in the leadership group of a construction team of 55
- Skilled public speaker and presenter, engaging in industry events, podcasts, community forums, client briefings, stakeholder meetings, toolbox talks, and inductions

## Industry Leadership Roles

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### Committee Member

#### Australian Water Association (AWA) | Victoria 2024 - 2026

- Victorian Branch Committee Member contributing to industry standards and best practices
- Participate in technical committees on data management
- Networking and knowledge sharing across water industry
- Advocate for innovation in environmental compliance

### Founder

#### Beyond Zero – Australian Water Association National Sustainability Specialist Network 2018 - Present

- Lead sustainability initiatives in construction industry
- Develop frameworks for carbon-neutral infrastructure projects
- Engage with government and industry stakeholders
- Drive adoption of environmental best practices

### Lead Applicant

#### Diagnostic Environmental Reporting Tool – CRC-P Grant 2025

- Applying for federal grant to build relational and geospatial database platform
- Store, analyse and report on federal environmental contamination and biodiversity data
- Establish industry working group for environmental data management

## Prior Member

### Low Carbon Concrete Working Group – MTIA

- Contributed to development of low-carbon construction materials and practices
- Industry collaboration on sustainability standards

## Professional Experience

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### Senior Environmental Advisor

#### Clough (Clough Salini Joint Venture) | Snowy Hydro 2.0 Project, NSW *January 2021 - July 2021*

- Senior advisory role with the managing contractor on the \$5.1 billion renewable energy project in NSW being delivered on behalf of the Federal Government and Snowy Hydro Ltd
- Implementation of the Biodiversity Management Plan – balancing construction needs with the conservation of sensitive flora and fauna within the National Park
- Key stakeholder management position – face-to-face coordination with the client (Snowy Hydro), impacted residents and inter-governmental agencies (NPWS, NSW-EPA, DPIE)
- Guide environmental approvals for construction teams to progress works within the Kosciusko National Park (KNP), including road and drainage design, erosion and sediment control planning and approvals, national park clearing permitting and risk mitigation for pollution and sensitive waterways
- Preparing, reviewing, and project managing the implementation of large-scale Erosion and Sediment Control Plans, drainage, and sediment basin construction as per Blue Book specifications
- Managing the ground and surface-water ESDAT chemical database for the Snowy 2.0 Project, required for the upkeep under the Environmental Protection Licence

### Environmental Coordinator

#### John Holland / Bouygues / Lendlease JV | Melbourne Metro Tunnel Project *January 2018 - December 2020*

- Work with key stakeholders including the City of Melbourne, EPA-Vic and Heritage Victoria to ensure planning approval and community consultation prior to staged construction works; ongoing engagement with impacted stakeholders in the CBD including presentations to community reference groups
- Managed environmental monitoring systems, including telemetered noise, air, and dust monitors across CBD sites
- Project managing large groundwater, surface-water and soil contamination monitoring events
- Successful approvals and excavation of twelve Heritage Inventory archaeological digs, the biggest archaeological investigation in Victoria's history

### Environment Engineer

#### WSP | Melbourne, Australia *2016 - 2018*

- Environmental engineering and consulting
- Project management and stakeholder engagement
- Environmental impact assessments and approvals

### Environmental Technician

#### Enqip | Melbourne, Australia *2015 - 2016*

- Environmental monitoring and field work
- Data collection and reporting
- Contaminated land investigations

### Geologist

#### Minerals and Metals Group (MMG) *2011 - 2013*

- Geological field work and exploration
- Data analysis and geological mapping
- Mining industry experience

## Education & Professional Development

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### Executive Master of Business Administration (EMBA)

#### RMIT University | Melbourne, Australia *2023 - 2025*

- Corporate Finance & Corporate Accounting
- Creating Business Strategies & Innovation in a Global Context
- Design Thinking & Marketing
- Ethical Leadership and Business Research Design
- Managing Business Operations & Executive Consulting

## **Bachelor of Science (Honours) - Geophysics**

**University of Melbourne | Melbourne, Australia** 2014

- Honours degree in Geophysics
- Research and thesis completion

## **Bachelor of Science - Geology**

**University of Melbourne | Melbourne, Australia** 2009 - 2011

- Major in Geology
- Strong foundation in Earth Sciences

## **Technical Skills**

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- **Database Development:** PostgreSQL, SQL, cloud-based database systems, geospatial databases, ESDAT chemical database management
- **Programming:** Python, SQL query optimization, database design, data analysis
- **Environmental Systems:** ISO 9001 and 14001 accredited systems, environmental management plans, sustainability frameworks
- **Data Management:** Construction material tracking, carbon accounting systems, environmental monitoring data
- **Sustainability:** Carbon accounting, embodied carbon tracking, SBTi, National Greenhouse Energy Reporting Scheme
- **GIS & Spatial Analysis:** Geospatial database platforms, spatial data analysis
- **Regulatory Compliance:** EPAV, Heritage Victoria, Traditional Owner approvals, EPA licensing
- **Project Management:** Tender preparation, budget management, subcontract management, team leadership

## **Certifications & Training**

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- **ISO 14001 Environmental Management Systems** - Lead Auditor
- **Blue Book** - Erosion and sediment control training
- **Construction White Card**
- **First Aid / Mental Health First Aid Certificate**

## **Publications & Presentations**

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- **“Project Pathway to Net Zero”** - Oz Water 2025 Technical Paper
- Multiple industry presentations and conference talks
- Podcasts on sustainability and environmental management
- Community forums and stakeholder briefings

## **Professional Memberships**

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- Australian Water Association (AWA) - Victorian Branch Committee Member (2024-2026)
- Environment Institute of Australia & New Zealand (EIANZ)
- Engineers Australia

## **Referees**

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**James Le - Client Representative** +61 429 230 466 Senior Project Manager - Melbourne Water

**James Braham - Previous Manager** +61 418 915 754 National Environment and Sustainability Manager - Downer

# Dr. Mahdi AbuAli, PhD

GDAC Saudi Lead



**GDAC-SA Tender Role:** Senior Technical Advisor | 35+ Years Saudi Aramco | Fleet Tahreez Managing Director (KSA) | 50+ Prospects Risked | Native Arabic Speaker

**Senior Geological Consultant - Saudi Arabia Petroleum Expert**

## Current Positions

### COO and VP

**Raphael Energy Group | Australia (Remote / Australia & KSA)** January 2021 - Present (5 years) | Part-time

Provide corporate leadership across portfolio strategy, governance cadence, and AOP/budget ownership, working with directors and partners to align technical evaluation with investment decisions. Focused on operating discipline, HSE/ESG, and building digital capability.

**Key Achievements:** - Established board reporting and rhythm for AOP, performance, and governance - Integrated 3D grav/mag with IHI to prioritize opportunities and move prospects through internal stage-gates - Strengthened stakeholder engagement with directors, investors, and partners

Raphael Energy Group Limited is exploring for oil and gas in the Officer Basin, onshore Western Australia, using an innovative approach integrating 3D gravity/magnetic mapping with microseepage induced indirect hydrocarbon indications (IHIs) such as radiometric gamma anomalies, to provide 3D GravMag IHI leads.

### Managing Director

**Fleet Tahreez Company Limited | Riyadh, Saudi Arabia** July 2025 - November 2025 (5 months) | Full-time

Leading mineral exploration initiatives that apply satellite- and AI-enabled geoscience to generate, rank, and de-risk targets for tier-one discoveries. Bridge operator needs with advanced sensing, turning technical insight into decision inputs for work-program planning and stakeholder reviews across KSA.

**Key Achievements:** - Introduced satellite/AI workflows that improved speed of exploration, target generation and prioritization - Delivered decision inputs adopted in planning reviews across KSA

### Senior Advisor

**Axiom Exploration Group | Saskatchewan, Canada (Hybrid)** April 2021 - June 2025 (4 years 3 months) | Part-time

Expanded remit from hydrocarbons to minerals, reflecting demand and the energy transition.

**Key Achievements:** - Supported opportunity screening and BD engagements for regional mandates - Broadened scope to minerals exploration aligned with market priorities

## Recent Consulting Experience

### Consultant & Project Manager

**Halliburton Kuwait | Kuwait** February 2023 - October 2023 (9 months) | Full-time

Led a multidisciplinary team on Paleozoic/Triassic gas exploration, delivering a ranked prospect inventory with full risking and volumetrics to inform client decision-making and program planning.

**Key Achievements:** - Matured, ranked, and risked 10 prospects across ~3,000 km<sup>2</sup> with supporting volumetrics - Inventory adopted for the next stage of planning by KOC stakeholders

## Saudi Aramco - 35 Years 9 Months of Distinguished Service

April 1985 - December 2020

### Senior Geological Consultant

**Saudi Aramco - Upstream Advanced Research Center | Dhahran, Eastern, Saudi Arabia** June 2014 - December 2020 (6 years 7 months) | Full-time

Drove deployment of exploration technologies to lower risk, mentored junior geoscientists, and embedded new methods into daily workflows—strengthening technical quality and decision confidence. Published papers and authored patents.

**Key Achievements:** - Embedded risk-reducing technologies and practices across teams - Delivered internal courses and editorial leadership to raise technical standards

## Senior Geological Consultant

**Saudi Aramco - Strategic Planning & Optimization (Exploration) | Dhahran, Saudi Arabia** February 2011 - June 2014 (3 years 5 months) | Full-time

Member of a senior team standardizing portfolio risking and shaping the exploration drilling schedule. Partnered with leadership on investment cases, approvals, and AOP delivery.

**Key Achievements:** - Standardized prospect risking across the national portfolio; matured and risked 50+ prospects, several drilled and resulting in oil & gas discoveries - Prepared decision papers and presented to senior leadership to inform approvals and execution

## Senior Geological Consultant

**Saudi Aramco - Upstream Ventures (JV Programme Manager) | Dhahran, Saudi Arabia** September 2006 - February 2011 (4 years 6 months) | Full-time

Managed JV exploration programs (SRAK, EniRepSa, SSG, LukSar), representing Aramco in governance forums and partner committees. Oversight spanned large onshore areas with planned wells and development concepts progressed with JV partners.

**Key Achievements:** - Oversaw >200,000 km<sup>2</sup> of JV acreage; planned several exploration wells and advanced the Kidan Arab sour-gas development plan - Prepared decision papers and secured partner approvals for work programs and budgets

## Senior Geologist - Earlier Roles

**Saudi Aramco | Dhahran, Saudi Arabia** June 1989 - September 2006 (17 years 4 months) | Full-time

**Progression:** Geochemist → Reservoir/Exploration Geologist → Geological Specialist/Team Lead

Progressed from hands-on geochemistry into exploration and specialist leadership, supporting prospect generation through petroleum systems modeling, source-oil correlations, reservoir characterization (clastic & carbonate), and 2D/3D seismic interpretation.

Partnered with multi-disciplinary teams to translate subsurface insight into drillable opportunities and helped establish repeatable workflows adopted across exploration.

**Key Achievements:** - Built regional play-fairway and charge/migration frameworks used in exploration interpretation - Delivered geochemical correlations and subsurface interpretations that underpinned prospect maturation and investment decisions

## Geologist

**Saudi Aramco | Dhahran, Saudi Arabia** April 1985 - September 1996 (11 years 6 months) | Full-time

Geologist and Geochemist, worked on Exploration of Central Arabia, Northwest and Red Sea areas. Conducted R&D studies on the source rock and migration of hydrocarbons, established a new source rock for the Paleozoic of Saudi Arabia. Published and presented many articles on the origin of hydrocarbons in the Paleozoic petroleum systems.

## Core Expertise

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- **Regional Geology:** Deep expertise in Arabian Peninsula, Middle East petroleum systems
- **Exploration Strategy:** Portfolio management, prospect risking, resource assessment
- **Basin Analysis:** Petroleum systems modeling, play concept development
- **Leadership:** Team building, mentoring, stakeholder management, multicultural team leadership
- **Saudi Operations:** 35+ years working in Saudi Arabia with comprehensive understanding of local context
- **Technical Integration:** Multidisciplinary data integration and interpretation
- **Business Development:** Partnership negotiations, JV management, joint ventures
- **Emerging Technologies:** Satellite/AI-enabled exploration, 3D GravMag IHI methods
- **Minerals Exploration:** Recent experience expanding from hydrocarbons to critical minerals

## Technical Skills

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- **Exploration Management:** Project management, exploration portfolio analysis, budgeting, strategic planning
- **Geological Analysis:** Basin analysis, petroleum systems analysis, reservoir characterization, field work, sampling
- **Data & Analytics:** Data analysis, quantitative research, qualitative research, analytical skills
- **Technical Disciplines:** Seismic interpretation, log interpretation, geochemistry, source rock analysis
- **Leadership & Management:** Team management, coaching, multicultural team leadership, contractual obligations
- **Specialized:** AI exploration, satellite-enabled exploration, mineral exploration, HSE/ESG

## Education

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### PhD - Geosciences

**RWTH Aachen University | Germany** 2001 - 2005

- Advanced research in petroleum systems and basin analysis
- Published research in peer-reviewed journals
- Activities and societies: AAPG, SPE, EAGE, IMOG, ALAGO, AAPG

### Master of Science (MS) - Geochemistry

**Colorado School of Mines | Colorado, USA** January 1986 - May 1989

- **Grade: 3.2/4.0**
- Specialization in petroleum geochemistry

### Bachelor of Applied Science (BASc) - Industrial Chemistry

**King Fahd University of Petroleum & Minerals (KFUPM) | Dhahran, Saudi Arabia** September 1980 - January 1985

- **Grade: 3.0/4.0**
- Strong foundation in chemistry and petroleum sciences

## Languages

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- **Arabic:** Native speaker - critical for Saudi stakeholder engagement
- **English:** Fluent - international business and technical communication

## Professional Memberships

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- American Association of Petroleum Geologists (AAPG)
- Society of Petroleum Engineers (SPE)
- European Association of Geoscientists and Engineers (EAGE)

# Dr. Qusay Abeed, PhD

Geological Director



**GDAC-SA Tender Role:** GDAC Technical Director | Native Arabic Speaker | Halliburton Petroleum Systems Lead | CCUS Expertise (Vision 2030 Alignment)

**GDAC Technical Director - Petroleum Geochemist**

## Current Positions

### Petroleum Geochemist Senior / Permedia and DSG Petroleum Systems Lead QA

**Halliburton | United Kingdom** *March 2020 - Present (5 years)*

- Senior role overseeing petroleum systems quality assurance for Permedia and DSG products
- Expertise in C++ programming and Carbon Capture, Utilisation and Storage (CCUS)
- Technical leadership for petroleum systems modeling and interpretation

### Permedia CO2 Team Agile Scrum Master

**Landmark Software (Halliburton) | Oxford, England, United Kingdom (Hybrid)** *March 2017 - Present (8 years)*

- Dealing with & developing technologies for basin modelling and CO2 storage
- Leading Agile/Scrum practices for CO2 technology development team
- Technical expertise in C++ and Carbon Capture, Utilisation and Storage (CCUS)

## Core Expertise

- Carbon Capture & Storage (CCUS)
- Petroleum Geochemistry
- Basin Modelling
- Petroleum Systems Analysis
- Source Rock Characterization
- Oil-Source Rock Correlation
- C++ Programming
- Agile/Scrum Project Management

## Languages

- **Arabic:** Native speaker - Direct communication with Saudi Geological Survey stakeholders
- **English:** Fluent - International professional communication

## Key Professional Experience

### Petroleum Geochemist / Petroleum Systems Lead

**Exploration Insights, Landmark, Halliburton | Abingdon, UK** *October 2013 - March 2020 (6 yrs 6 mos)*

- Working as Petroleum System Lead, dealing with search, gathering, and processing petroleum systems data
- Global coverage: South America, Central America, North Africa, Lower 48 (USA) and Middle East
- **Created Global Petroleum Systems Interpretations Maps** for diverse regions
- Deep understanding of Middle East petroleum systems including Mesopotamian Basin
- Skills: Log Interpretation, Research and Development (R&D)

### Petroleum Geologist / Post-Doctoral Researcher

**RWTH Aachen University | Aachen, Germany** *October 2012 - October 2013 (1 yr 1 mo)*

- Post-doctoral research on source rock/oil characterization, oil-source rock correlation, and oil-oil correlation
- Focus on bulk properties, biomarkers and molecular geochemistry of oil
- **Basin modelling for the Mesopotamian Basin of Iraq** - one of the largest petroleum systems in the world
- Gained experience in managing petroleum system modelling and seismic interpretation

## Research Associate

**RWTH Aachen University | Aachen, Germany** *October 2008 - October 2012 (4 yrs 1 mo)*

- Petroleum systems research at Institute of Geology and Geochemistry of Petroleum and Coal
- Advanced research in organic geochemistry and thermal/burial history
- Skills: Research and Development (R&D), Presentations

## Researcher and Exploration Instructor

**Tikrit University | Tikrit, Iraq** *November 2006 - May 2008 (1 yr 7 mos)*

- Researcher at the Institute of Geology and Geochemistry of Petroleum and Coal at RWTH Aachen University
- Main field of interest: Organic geochemistry of petroleum and source rocks
- Thermal/burial history and basin modelling research
- Skills: Research and Development (R&D), Presentations

## Education

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### PhD - Petroleum Geochemistry & Basin Modelling

**RWTH Aachen University | Aachen, Germany** *2008 - 2012*

- **Grade: Magna Cum Laude**
- Thesis focused on Mesopotamian Basin petroleum systems

### MSc - Petroleum Geology

**University of Baghdad | Baghdad, Iraq** *2002 - 2006*

- **Grade: 81.91%**

### BSc - Geology/Earth Science

**University of Baghdad | Baghdad, Iraq** *1998 - 2002*

- **Grade: 71.7%**

## Technical Skills

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- **CCUS Technologies:** Carbon capture, utilisation and storage systems - aligns with Saudi Vision 2030 sustainability initiatives
- **Programming:** C++ for petroleum systems modeling and basin simulation
- **Geochemical Analysis:** Biomarkers, molecular geochemistry, source rock characterization
- **Basin Modelling:** Thermal history, burial history, petroleum systems modeling
- **Seismic Interpretation:** 2D/3D seismic data interpretation
- **Log Interpretation:** Well log analysis and correlation
- **Project Management:** Agile/Scrum methodologies

## Geographic Expertise

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- **Middle East:** Mesopotamian Basin (Iraq), North Africa - deep regional knowledge
- **Americas:** South America, Central America, Lower 48 (USA)
- **Europe:** UK, Germany - extensive work experience

# Moritz Theile

Chief Technology Officer



**GDAC-SA Tender Role:** Chief Technology Officer | 6+ Years Lithodat CTO | Full-Stack Architect

Chief Technology Officer

## Current Positions

### Chief Technology Officer (CTO)

**Lithodat Pty Ltd | Melbourne, Australia** 2019 - Present (6+ years)

- Architect and Lead Developer for all Lithodat technical infrastructure
- Building IT Infrastructure from ground up
- Developer Onboarding and Team Building
- Responsible for all technical architecture and system design decisions

### Self-Employed Software Architect

**MTheile.com | Greater Munich Metropolitan Area, Germany** 2011 - Present (14 years)

- Delivering IT solutions that actually work
- Starting with analysis, ending with thought-through systems
- Multiple successful project deliveries across diverse industries

## Core Expertise

- Full-Stack Development
- Cloud Architecture (AWS)
- Java / Spring Boot
- JavaScript / TypeScript / React
- SQL / Database Design
- Data Analytics (Tableau, ELK)
- Agile Methodologies
- System Architecture

## Key Projects at MTheile.com

### HotelOrga

#### Hotel Management Platform

- Comprehensive platform for hotel operations management
- Full-stack development with modern architecture

### TeamTable

#### Generic Database System

- Flexible database system for team collaboration
- Custom data modeling and analysis capabilities

### MonaStack

#### Technology Stack Development

- Modern technology stack for rapid application development
- Reusable components and architecture patterns

### Cutworks

#### Laser-Cut Metal Parts Ordering System

- E-commerce platform for manufacturing industry
- Integration with manufacturing systems and logistics

## Docsafe

### Consulting and Development for Swisscom

- Enterprise-level document management solution
- Security and compliance requirements

## Professional Experience

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### Project Manager

**4Soft | Germany** 2007 - 2011 (4 years)

Led multiple enterprise software projects for financial services industry:

#### DynSBatch

##### Bulk Data Processing Software

- High-performance batch processing system
- Enterprise-scale data transformation and validation

#### BeteiligungsIntegrator

##### Interface for Investment Funds

- Integration platform for financial institutions
- Complex data synchronization and validation

#### DWH4OSPlus

##### Framework for Data Analysis

- Data warehouse framework for business intelligence
- Analytics and reporting capabilities

#### WAC Manager

##### Portfolio Management Software

- Investment portfolio tracking and analysis
- Real-time market data integration

### Java Developer

**Macquarie University | Sydney, Australia** 2005 - 2006 (1 year)

Developer for Australian Higher Education systems:

#### MAMS (Meta Access Management System)

##### Australian Higher Education Platform

- Identity and access management for university systems
- Integration across multiple institutions

#### Autograph

##### Shibboleth Extension

- Enhanced authentication capabilities
- Single sign-on infrastructure

## Education

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### Computational Science

**Technical University of Munich (TUM) | Munich, Germany** 1998 - 2004

- Focus on computational methods and software engineering
- Strong foundation in mathematics and computer science

### Exchange Student

**Hoover Highschool | Fresno, California, USA** 1994 - 1995

- International education experience

### Highschool Graduation (Abitur)

**Adalbert-Stifter Gymnasium | Germany** 1987 - 1997

## Technical Skills

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- **Backend Development:** Java, Spring Boot, SQL, Maven, REST APIs
- **Frontend Development:** JavaScript, TypeScript, React, HTML5, CSS3
- **Cloud & Infrastructure:** AWS (EC2, S3, RDS, Lambda), Docker, CI/CD
- **Database:** SQL (PostgreSQL, MySQL), NoSQL, Data Modeling
- **Data Analytics:** Tableau, ELK Stack, UML, Data Visualization
- **Methodologies:** Agile, Scrum, Kanban, DevOps practices
- **Architecture:** Microservices, Distributed Systems, System Design

## Professional Skills

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- **Leadership:** Team building, mentoring, technical direction
- **Problem Solving:** Complex system debugging and optimization
- **Communication:** Technical documentation, stakeholder management
- **Creativity:** Innovative solutions to technical challenges

# Dr. Behnam Sadeghi

ML Technical Advisor



**GDAC-SA Tender Role:** Technical Advisor | First Australian IAMG Vistelius Award | Fulbright Scholar | 58+ Publications | 3 Python Packages | JORC Certified

Principal Consultant & Research Fellow PhD, FAIG, FAusIMM, FAAG, RPGeo, Fulbrighter

## Major Awards & Recognition

- **2023 IAMG Vistelius Award** - First Australian recipient
- **2022-2024 Fulbright Scholar** (USD 29,325)
- **2020 IAMG Founders Award** - First Ever
- **FAusIMM** (Fellow of Australasian Institute of Mining and Metallurgy)
- **FAAG** (Fellow of Association of Applied Geochemists)
- **FAIG** (Fellow of Australian Institute of Geoscientists)
- **RPGeo 10295** (Registered Professional Geoscientist)

## Current Positions

### Research Fellow - Critical Minerals Exploration

**CSIRO (Australian Resources Research Centre) | Australia 2023 - Present**

- Leading research on critical minerals exploration methodologies
- Focus on rubidium, lithium, and rare earth elements

### Visiting Research Fellow

**UNSW - Dept. of Biological, Earth and Environmental Sciences 2022 - Present**

### Principal Consultant

**Independent Practice 2022 - Present**

## Research Fellowships

- **Fulbright Research Scholar** - Carnegie Institution for Science, Earth and Planets Laboratory + Stanford University, USA (2023-2024)
- **Honorary Research Fellow** - EarthByte Group, University of Sydney (2022-2023)

## Core Expertise

- Fractals & Multifractals
- Exploration Geochemistry
- Uncertainty Quantification
- Machine Learning in Geosciences
- GIS & Data Analytics
- Critical Minerals
- Python Development
- JORC Code Reporting

## Publications Summary

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- **58+ peer-reviewed journal articles** in leading geoscience journals
- **34+ conference presentations, invited lectures and keynote speeches** (oral and poster)
- **2 books** including Elsevier reference book on Fractals and Multifractals
- **8 book chapters** in Encyclopedia of Mathematical Geosciences
- **3 Python packages** developed for geoscience applications

## Key Publications

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### “Fractals and Multifractals in the Geosciences”

#### Elsevier Reference Book, 2024

- Comprehensive reference work on fractal applications in geosciences
- ISBN: 978-0-323-90897-9

### Selected Journal Articles

- Sadeghi, B., Cohen, D.R. (2023) “Decision-making within geochemical exploration data based on spatial uncertainty” - Ore Geology Reviews
- Sadeghi, B. (2022) “Chatterjee Correlation Coefficient: A robust alternative for classic correlation methods” - Ore Geology Reviews
- Sadeghi, B. (2021) “Simulated-multifractal models: a futuristic review” - Ore Geology Reviews

## Python Packages Developed

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- **pyClusterWise** - Clustering method selection with uncertainty analysis
- **IRWPy** - Inverse Radius Weighting interpolation for spatial data
- **TripleCpy** - Chatterjee Correlation Coefficient implementation

## Editorial & Leadership Roles

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### Associate Editor (3 journals)

- Ore Geology Reviews
- Geochemistry/Chemie der Erde
- Ore and Energy Resource Geology

### Editorial Board Member (5 journals)

- Applied Computing and Geosciences
- Earth Science Informatics
- Minerals
- Mining
- Discover Minerals

### Professional Society Leadership

- AAG Councillor (2024-2025)
- AAG Strategic Planning Team (2017-2027)
- IAMG Outreach Committee Member (2023-Present)
- IAMG Liaison Committee Member (2023-Present)
- AIG Official Assessor for RPGeo applications

## Key Professional Experience (2007-Present)

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### Senior Geologist & Consultant

Perpetual Resources Ltd (ASX: PEC) | Australia 2022

### Senior Geologist & Data Analyst

Impact Minerals Ltd (ASX: IPT) | Australia 2021 - 2022

## Researcher / Technical Software Supporter

EarthByte Group, University of Sydney | Australia 2020 - 2021

- GPlates software tutorials and technical support

## Senior Geologist & Data Analyst

Satrap Resources (WA) | Western Australia 2020 - 2021

## Senior Geologist

Hampa Behineh Consultant | Iran 2014 - 2020 (6 years)

## Researcher

James Cook University - Economic Geology Research Unit (EGRU) | Australia 2015 - 2017

## Education

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### Ph.D. of Geosciences

UNSW, Australia 2018 - 2020

- Thesis: "Quantification of Uncertainty in Geochemical Anomalies in Mineral Exploration"

### Master of Mining Engineering - Exploration

Islamic Azad University (IAU), Iran 2009 - 2012

### Bachelor of Mining Engineering - Exploration

University of Kurdistan (UoK), Iran 2004 - 2009

## Recent Certifications (2024-2025)

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- Learn to Lead with UNSW 2025: Strategic Thinking for Impactful Leadership
- Standard Mental Health First Aid - Australian Red Cross
- **Professional Certificate in JORC Code Reporting - AusIMM**
- Using GenAI Effectively at Work - CSIRO
- Python for Geoscientists - Curtin University
- Regolith Geology and Geochemistry - CSIRO
- PMP Certification Exam Prep Course - Udemy

## Teaching & Mentoring

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- IAMG Short Course "Applied Geochemistry and Analytics" (August 2023)
- GEOL 5806 Graduate Course in Exploration Geochemistry - Laurentian University (December 2022)
- Co-advisor on Geology-Machine Learning projects - UNSW (2021-Present)
- GPlates software tutorials provider - EarthByte Group (2020-2021)

## Professional Society Memberships

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- Australian Fulbright Alumni Association (AFAA)
- The American Geophysical Union (AGU)
- The Geological Society of America (GSA)
- AusIMM - Fellow (FAusIMM)
- AIG - Fellow and Registered Professional Geoscientist (FAIG, RPGeo)
- European Geosciences Union (EGU)
- Society for Geology Applied to Mineral Deposits (SGA)
- IAMG - Lifetime Member
- AAG - Fellow (FAAG)

# Vinko Novak

## Head of Data Security

Digital Transformation | Domain-Driven Design | Enterprise Architecture

+49 151 11 44 68 77 | vinko.novak@scenario.de



- **Digital Transformation Expertise:** 18+ years leading digital platform development and modernization initiatives
- **Strategic Consulting:** Proven ability to align technology with business strategy at C-level (GEMA, Deutsche Bahn, BMW)
- **Domain-Driven Design:** Expert in DDD, Event Storming, and strategic architecture - essential for complex geological data platform design
- **Enterprise Architecture:** Architecture reviews for major organizations (Deutsche Bahn, BMW, GEMA) - brings enterprise-scale expertise
- **Agile Leadership:** 10+ years as Tech Lead with Scrum Master and Product Owner experience - effective team coordination
- **Complex Integrations:** Successfully integrated systems across multiple organizations in joint ventures - critical for GDAC data platform
- **Visualization Skills:** Ability to translate complex technical concepts for executive decision-making - bridges technical and business stakeholders
- **International Experience:** Collaboration with teams across Germany, UK, USA, Tunisia, Hungary, Romania - global perspective
- **Startup & Innovation:** Experience taking startups from concept to successful funding rounds - entrepreneurial mindset
- **Cloud & Modern Tech:** Expertise in Kubernetes, Docker, Terraform, microservices - modern infrastructure capabilities for scalable platform



# Dr. Romain Beucher

## Geoscience Modelling & Software Engineering Expert

Thermo-Mechanical Modelling | Surface Processes | Research Software Development

Canberra, Australian Capital Territory, Australia



**Professional Summary:** Dr. Romain Beucher is a geoscientist and research software engineer with extensive expertise in lithospheric-scale thermo-mechanical modelling, surface processes, and low-temperature thermochronology. With over 15 years of research experience across leading institutions in Australia, Norway, and the UK, he specializes in coupling large-scale basin models with surface processes to study interactions between erosion and tectonics. Currently leading the Model Evaluation and Diagnostics team for the Australian Earth-System Simulator National Research Infrastructure (ACCESS-NRI), Dr. Beucher combines deep geological knowledge with advanced software engineering capabilities.

## Professional Experience

### Team Leader (Senior Manager) – Model Evaluation and Diagnostics

#### ACCESS-NRI (Australian Earth-System Simulator National Research Infrastructure)

August 2022 – Present (3+ years) | Canberra, ACT, Australia

Leading the development and evaluation of model diagnostics for Australia's national Earth-system modelling infrastructure.

**Key Skills:** Negotiation, Training, Team Management, Release Management, Robust Design, Critical Thinking, Problem Solving, Teamwork

### Researcher

#### The Australian National University (ANU)

February 2020 – August 2022 (2 years 7 months) | Canberra, ACT, Australia

Conducted advanced research in lithospheric-scale thermo-mechanical modeling of rifts and passive margins. Specialized in surface processes modeling with focus on quantifying rock exhumation and relief evolution using low-temperature thermochronology (apatite Fission Track, U-Th/He). Developed coupled large-scale basin models integrating surface processes to study erosion-tectonic interactions and feedbacks.

**Key Skills:** Training, Release Management, Robust Design, Critical Thinking, Problem Solving, Teamwork

### Research Software Engineer

#### AuScope

October 2019 – August 2022 (2 years 11 months)

Developed research software infrastructure for geoscience applications.

**Key Skills:** Training, Release Management, Robust Design, Critical Thinking, Problem Solving, Teamwork

### Visiting Research Fellow

#### The Australian National University (ANU)

October 2019 – February 2020 (5 months) | Canberra, Australia

**Key Skills:** Critical Thinking, Problem Solving

### Postdoctoral Researcher

#### University of Melbourne

October 2016 – February 2020 (3 years 5 months) | Melbourne, Australia

**Key Skills:** Critical Thinking, Problem Solving

### Postdoctoral Researcher

#### University of Bergen (UiB)

June 2013 – June 2016 (3 years 1 month) | Bergen Area, Norway

**Key Skills:** Critical Thinking, Problem Solving

### Research Associate

#### University of Glasgow

June 2010 – June 2013 (3 years 1 month) | Glasgow, United Kingdom

**Key Skills:** Critical Thinking, Problem Solving

## Education

### Doctor of Philosophy (PhD) – Geology/Earth Science

Université Grenoble Alpes

2006 – 2009

**Master of Science (MS) – Geology/Earth Science**

**Université Grenoble Alpes**

2004 – 2006

**Bachelor of Science (BS) – Geology/Earth Science**

**Le Mans Université**

2001 – 2004

**Core Competencies**

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- Lithospheric-scale thermo-mechanical modelling of rifts and passive margins
- Surface processes modeling and geomorphological evolution
- Low-temperature thermochronology (apatite Fission Track, U-Th/He dating)
- Quantitative analysis of rock exhumation and relief evolution
- Coupled basin-scale models integrating tectonics and erosion processes
- Research software engineering and model development
- Earth-system model evaluation and diagnostics
- Team leadership and research project management

# Annemarie Grass

Head of Cyber Defence

**Email:** annemariegrass123@outlook.com

An experienced leader with over 10 years leading both operational and technical teams across cyber domains. Specialist in cyber incident response, threat hunting, and implementing detection and response capabilities. I am passionate about building, operating, and enhancing organisations defensive security, responding to security incidents, and uplifting overall security maturity.

## Professional Experience

- Head of Cyber Defence | IDP Education Pty Ltd | June 2023
- Senior Manager, Defensive Security | PricewaterhouseCoopers (PwC) | July 2022
- Manager, Defensive Security | PwC | June 2021
- Cybersecurity Lead Forensics & Information Intelligence | AGL | January 2020
- Cybersecurity Lead 3rd Party Security Management | AGL | October 2018
- Lead Financial Crime AML/Sanctions | NAB | May 2018
- Senior Business Analyst | NAB | March 2016
- Senior Change Analyst September | NAB | September 2015

## Training and Education

- Foundations of Directorship | Australian Institute of Company Directors | 2024
- Charles Sturt University | Master of Information Technology; Cybersecurity | 2020
- Nuix | Workstation Data Discovery Specialist | 2019
- Adaptive Frameworks | PRINCE 2 – Foundation and Practitioner | 2014
- Griffith University | Bachelor of Business; Mgr. Management and Human Resources | 2013

## Demonstrated Skills and Strengths

- Ability to effectively bridge the gap between heavily technical environment and business strategy requirements to deliver outcomes to an organisation, achieved through balancing communication styles across stakeholders at all levels.
- Collaborative approach to problem solving, with ability to engage teams and bring them on the journey to achieve outcomes for organisations.
- Experienced Incident Responder coordinating activities for major multi-national organisations and global operations. Routinely manages multiple teams of investigators spanning across technology, internal audit and business assurance functions, profiles threat actors, works closely with senior executives on sensitive investigations and leading meaningful Post Incident Reviews.

## Employment History

### IDP Education Ltd

Head of Cyber Defense

Dimensions and Key Achievements

- Responsible for the establishment of the Security Operations Centre, including development of technical incident response plans, playbooks and training and onboarding of new security operations analyst.
- Responsible for leading the establishment of a Cyber Deliver Centre in Manila from initial market scan, vendor partnering and onboarding of 18 experienced cyber security practitioners to support IDP in delivery of it's Cyber Uplift Strategy.
- Responsible for the establishment of a Global Crisis Management Framework for IDP Education, working and engaging with senior executives to develop and refine the framework.
- Accountable for the delivery of several global incident response activities, working with regional technology teams to improve detection, response and recovery capabilities for regional technology teams responsible for local technology infrastructure.
- Led the global rollout of new Endpoint Detection and Response toolset for Servers and Workstations.
- Development of detection rule sets for cloud-based infrastructure assets, onboarding these into IDP Security Incident Event Monitoring platform.
- Established forensics capabilities to support whistle blower investigations globally.
- Presented on the progress against the Cyber Uplift Program to the IDP Board and led a board level incident response exercise focusing on key decisioning around recovery and response.
- Delivered NIST Maturity Uplift across the following domains (maturity uplift results determined by external vendor audit)
  - Protect 1.90 (2022) – 2.7 (2025)
  - Detect 1.30 (2022) – 3.2 (2025)
  - Respond 2.08 (2022) – 3.6 (2025)

## **PwC**

### **Senior Manager Defensive Security July 2022**

#### **Dimensions and Key Achievements**

- Security Operations secondment to a multinational mining company for a five-month period reporting directly to the Global Vice President, with responsibilities that include
  - Identifying and delivering on opportunities to uplift the company's Cyber Threat Intelligence function, Security Operations – Security Orchestration Automation Response (SOAR) tooling implementation, detection, and response capability.
- Development of operating models for incident response capabilities, where the capability is responsible for servicing external and internal client stakeholders, including government agencies.
- Incident responder responsible for leading client engagements for organisations large and small to detect, contain, and recover from cyber security incidents. Responding to several major ransomware incidents, as well as incidents involving business email compromise and accidental data exposure.
- Development of thought leadership pieces with respect to security operations, ransomware in the supply chain.
- Delivery of presentations to board members on contemporary cyber threats organisations are facing, bringing real world experience to the conversation to improve organisations preparedness to response to cyber incidents.
- Member of the 'She's Here' in Technology working group and was responsible for the coordination and delivery of PwC International Women's Day campaign for 2023. In addition to this I mentor female STEM graduates across PwC globally supporting females in reaching their full potential in the workforce.

## **PwC**

### **Manager Defensive Security June 2021**

#### **Dimensions and Key Achievements**

- Led engagements reviewing Cybersecurity submissions to regulatory bodies to validate funding requirements to uplift Cyber maturity in line with regulatory frameworks.
- Delivery of post incident reviews for major ransomware incidents, supporting clients in understanding the value of reflection and helping clients to identify and capitalise on opportunities to improve capability and build resilience after a cyber incident.
- Led the development and delivery of cyber crisis simulations for top executives and boards at several ASX10 and other major organisations.
- Coaching and mentoring junior team members on development of client engagements, this included 4 months in PwC Skilled Services Hubs developing our team's defensive security capability through the development and delivery of internal training.

## **AGL Energy**

### **Cybersecurity Lead – Forensics & Information Intelligence January 2020**

#### **Dimensions and Key Achievements**

- Leading a team of forensic investigators and cyber security engineers.
- Delivery of eDiscovery infrastructure (Nuix) on AGL Azure platform, reducing reliance on external vendors and reduction in data leaving the AGL environment
- Leading the delivery of eDiscovery RFP identifying best in breed external suppliers to work on AGL Nuix infrastructure to continue to develop eDiscovery capability at AGL
- Implementation of Azure Information Protection for all AGL users
- Responsible for both CASB DLP, Zscaler DLP infrastructure and ongoing policy development, refinement
- Developing information intelligence capability at AGL utilising industry leading technologies, OSINT and threat hunting
- Lead forensics investigations (including fraud, code of conduct violations) in coordination with internal teams, associated 3rd parties and government agencies.
- Collection and analysis of digital evidence in case of regulator investigation, legal request, or insider investigation
- Identify and analyse key pieces of intelligence information to provide input to business led security investigations
- Conduct forensics investigations into cyber related incidents within AGL.

## **AGL Energy**

### **Cybersecurity Lead 3rd Party Security Management October 2018**

#### **Dimensions and Key Achievements**

- Development of AGL strategy for management of existing and new 3rd party vendors
- Assessing and reporting on the security posture of AGL security relative to data security and fraud prevention controls
- Lead internal investigations on suspected or actual internal fraud activities this includes coordinating examining of technology pertinent to the investigation, stakeholder engagement, remediation strategy and reporting close out.
- Ensure that Security Incidents relating to data loss are captured in accordance with the Security Incident Management process. Monitor for Root cause analysis and remediation actions and assigned to owners to close the Incident with evidence.
- Provide reports to AGL's Privacy Officer where the security of AGL's data is weak or presenting risk of breach, work with the Privacy Officer and the business on treatment plans.

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# Nilesh Khadse

GIS Full Stack Developer



## Experience

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### Senior Full Stack Engineer (GIS)

#### Deimos Space UK · Full-time

Jun 2021 - Present · 4 yrs 7 mos

Harwell Campus, England, United Kingdom

- Developed numerous Earth Observation applications, involving the creation of APIs, user interfaces, and processing large volumes of satellite data.
- Collaborated with the European Space Agency to successfully deliver several projects, including:
  - Designed a space debris collision risk assessment web application for satellite operators.
  - Implemented an Authentication and Access Management (IAM) service using Keycloak OpenID and an Accounting system for a microservices platform catering to Earth Observation application providers. This microservice plays a central role in the EU's Destination Earth (DestinE) initiative.

**Skills:** Python, docker and +4 skills

### Senior MetOcean Software Developer

#### Fugro · Full-time

Nov 2019 - May 2021 · 1 yr 7 mos

Oxfordshire, England, United Kingdom

- Successfully implemented the Apache Airflow ETL platform for data processing workflows.
- Developed an atmospheric lightning data harvester in Python using the asyncio library, achieving a significant reduction in the server's CPU usage from 25% to 0.1% compared to its predecessor.
- Transformed the data ingestor application from a monolith Windows service to a containerized .NET Core application. This initiative not only reduced reliance on an aging data ingestion system but also made the team platform agnostic.
- Set up Elastic (ELK) analytics platform for monitoring workflows and applications, contributing to improved visibility and performance tracking.

**Skills:** Geoprocessing, Elastic Stack (ELK) and +3 skills

### Principal Software Developer

#### Neftex Exploration Insights · Full-time

Jun 2014 - Oct 2019 · 5 yrs 5 mos

Oxford, United Kingdom

- Ideated and developed the 'QuickPlates' web mapping application, showcasing reconstructed data through geological times. This application also provided a client data processing service, marking a notable achievement.
- Devised a geological data reconstruction algorithm based on scientific literature, enabling the provision of 'reconstruction as a service' for client data.
- Developed a tool utilizing the OpenCV library to extract geographical contour features from historical map images.
- Built an aesthetic geological timescale using D3.js.
- Developed numerous ArcGIS plugins, web mapping applications and geospatial ETL workflows using FME.

**Skills:** React.js, .NET Framework and +3 skills

### Senior GIS Analyst

#### HR Wallingford (SeaZone Solutions)

Apr 2013 - Jul 2014 · 1 yr 4 mos

#### SeaZone Solutions

4 yrs 7 mos

### GIS Analyst

Jan 2009 - Mar 2013 · 4 yrs 3 mos

### GIS Technician

Sep 2008 - Dec 2008 · 4 mos

# Education

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**The University of Edinburgh**

**MSc, GIS**

*2007 - 2008*

**K. K. Wagh Institute of Engineering Education**

**B. Tech, Agricultural Engineering**

*2003 - 2007*